

# **Understanding Crime in London: A Regression Analysis of Socio-economic Factors Across Regions from 2001–2013**

**Roshan Kerai**

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Department of Mathematics

Brunel University London

Supervisor: Dr Keming Yu

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# Abstract

This research paper examines the spatio-temporal trends of crime across the five regions of London between 2001 and 2013. It particularly focuses on the socio-economic factors that significantly influence crime. Using data merged from individual boroughs, a series of regression methods (such as **Poisson regression**, **Negative Binomial regression** and **Beta regression**) have been applied to investigate the relationship between crime and the various predictors, including population size, median house price, average income, percentage of jobseeker's allowance claimants, and job density.

The research begins with a descriptive analysis of crime over time, providing evidence of variations across different regions at different time periods. Overdispersion was detected in the crime count data, and to address this, Negative Binomial regression was used after Poisson regression to provide more accurate and significant estimates.

In order to make sure model assumptions were met, diagnostic checks such as VIF for multicollinearity, Q-Q plots for distribution, and residual plots were performed. To visually identify spatial patterns in crime across different boroughs, choropleth maps and line graphs were presented.

Therefore, the results provide an understanding of the distribution of crime in London, emphasising the importance of considering **temporal events** (like the 2008 financial crisis) and **local socio-economic factors**, which contribute to shaping crime patterns.

# 1 Introduction

Crime has been a subject of concern for the public for a long time as well as being a key subject to undertake research on, especially in more urban areas, where it is easier to identify socio-economic differences. Looking particularly into Greater London, the main root of crime is associated with the urban expansion and industrialisation that took place during the 19th and 20th centuries. This industrialisation led to the city transforming into a busy metropolis through the construction of new industries and advancements in technologies, leading to population growth and eventually changing the way societies behaved within London. Due to the vast amount of job opportunities available in the city, rural to urban migration became popular. As a result of this sudden surge in population into the city, supply of housing was inadequate as there were not enough houses to for everyone. Also, there was a lack in public health infrastructure like clean water and sewage systems, leading to further spread of diseases and overall, a poor standard of living in certain boroughs. Poverty and crime go hand in hand. Often, poverty led individuals to commit certain crimes because that was the only way to survive or to acquire goods and services that can be obtained illegally. However, over time changes in legislation and enforcement of stricter policing have definitely made individuals think twice before they engage in criminal activity.

Policy reforms implemented by the government influences the trend of crime and decisions of individuals on whether to engage in criminal activity or not. The 'Crime and Disorder Act 1998' was aimed to prevent crime by giving local authorities responsibility for preventing crime whilst working alongside the police. It also included new offences that individuals could be charged for. For example, a key policy from the act was section 34 of 'lowering the age of criminal responsibility to 10 years', meaning that children aged 10+ could be convicted of a crime, without the prosecution having to prove to the child that they did something seriously wrong. However, the penalties were not as strict as those for adults as age and maturity would be considered. The age of criminal responsibility being broadened meant that there was a higher chance of fluctuations in the crime rate. The 'Police and Justice Act 2006' was created to reform the UK criminal justice system and policing structure. It made key changes to how police authorities were chosen and how they were supervised. It also made changes to the bail system and the power that the police had over these bail conditions. This could indirectly affect crime rates as depending on whether they are released on bail or kept in custody, this may influence whether the individual would want to engage in criminal activities in the future or not. Also, in 2004, the Met police introduced 'Safer Borough Teams' whose core aim was to work with local communities and

residents to identify key issues within those specific boroughs and think of methods for how those issues could be resolved. However, even though all these policies and teams have been implemented, crime is not distributed evenly across the whole of London. There are certain boroughs that are crime hotspots because of high footfall and various socio-economic disparities between boroughs.

A major financial crisis occurred in 2008 leading to economic hardship and increases in the cost of living. Unemployment rates rose rapidly resulting in many individuals to claim jobseeker's allowance (unemployment benefit), especially in vulnerable areas. As a result, the incentive to commit crime would have increased, as individuals who are facing economic hardships would be desperate to become financially stable and may be tempted to engage in criminal activities to provide for themselves and their families. Therefore, interests have sparked into understanding which socio-economic factors drive crime.

In contrast, cities like Dubai have different methods. The UAE have more stricter laws that follow the Islamic (Sharia) law, which has a large impact on the low crime rates. The different policies in Dubai particularly focus on deterrence, with severe penalties for theft and other violent crimes. The implementation of high end security surveillance and strict action by police discourages individuals from even thinking about engaging in criminal activity leading to an overall sense of complete safety for all members of the public.

Understanding crime in London requires a perspective that looks into the different factors, such as socio-economic factors and government policies. This particular research looks into the crime trends within the five regions of London between 2001 and 2013 and compares trends between regions. The key methods used are Poisson, Negative Binomial and Beta regression to identify how certain factors like average income, house prices, job density, jobseeker's allowance claimants and population rates correlate and influence crime and also by how much strength.

## **2 Description of Data**

### **2.1 Data Sources**

The main dataset that has been used within this study is a combination of five other datasets. All of the datasets that were used to create this main dataset were sourced from the 'London





so I computed the same process for that data as well. Therefore, I had a dataset which included the population and crime count for each borough between 2001 and 2013.

### Geographical Partitioning

Greater London is built up of 32 boroughs (local districts) plus the City of London, each having its own governing council [Wikipedia contributors, 2025]. Each of the boroughs are divided into electoral wards in order to elect councillors. Undertaking regression analysis for every single London borough would be difficult so I decided to partition the 32 boroughs into 5 different sections, which was motivated by the figure below obtained through [Wikipedia contributors, 2024].

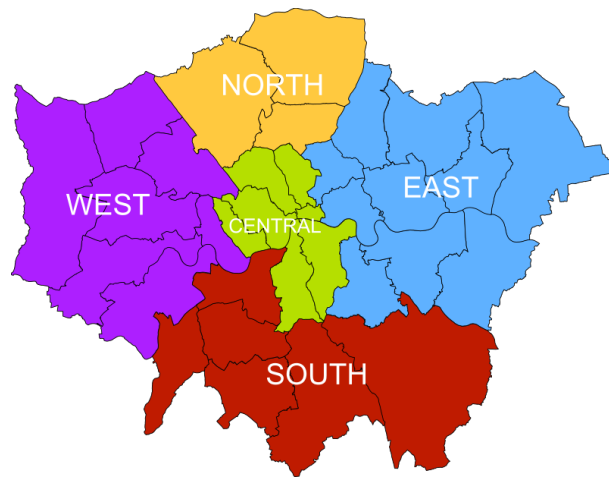


Figure 2: 5 sub-regions of Greater London

So, I decided to partition the 32 boroughs into the five sub-regions. In order to do this, I assigned each borough to the region that it fell into and then once again computed the average of all the boroughs within that particular sub-region (using the filter function in excel). This provided me crime counts for the five regions between the years 2001 and 2013. Once again, I applied the same process to the population data.

Therefore, after all the alternations, I had my new altered crime data file. Moving onto the files for the socio-economic factors, I had a file for each of the four different socio-economic factors that were going to be included. The median house price file already had the house prices for each borough for each year, so all that was needed to do was assign the boroughs to the sub-region and take the average. For the jobseeker's claimants rate, the values were given by borough but for each quarter within the years. So, I had to take the average of the four values to get a singular value for the specific year and then apply the same process of assigning the boroughs their sub-region and then averaging. The income file had the median and mean income for each year and borough, so I extracted the median income for each year and borough and applied the standard

process of assigning and averaging. For the last file of job density, the job density was already provided for each year, so I once again assigned and averaged.

After all these computations, I then had to combine all five of the files into one dataset, which was my final and main dataset that I had used to conduct my analysis.

| Region  | Y = No. of total crimes per year | X1 = Year | X2 = Population | X3 = Jobseeker claimants (%) | X4 = Median House Price (£) | X5 = Average Income level (£) | X6 = Job Density |
|---------|----------------------------------|-----------|-----------------|------------------------------|-----------------------------|-------------------------------|------------------|
| Central | 253396                           | 2001      | 1276929         | 3.5                          | 248167                      | 20817                         | 1.71             |
| Central | 261013                           | 2002      | 1278340         | 3.5                          | 277500                      | 21483                         | 1.62             |
| Central | 249325                           | 2003      | 1285046         | 3.4                          | 283833                      | 21850                         | 1.62             |
| Central | 242128                           | 2004      | 1298248         | 3.1                          | 306333                      | 21550                         | 1.6              |
| Central | 223887                           | 2005      | 1311560         | 3                            | 325833                      | 24317                         | 1.61             |
| Central | 221950                           | 2006      | 1324069         | 3                            | 360000                      | 24867                         | 1.58             |
| Central | 216253                           | 2007      | 1335472         | 2.5                          | 419167                      | 26733                         | 1.64             |
| Central | 207659                           | 2008      | 1346879         | 2.3                          | 435833                      | 27867                         | 1.71             |
| Central | 202505                           | 2009      | 1367880         | 3.4                          | 429167                      | 29000                         | 1.69             |
| Central | 203347                           | 2010      | 1385793         | 3.4                          | 481667                      | 28567                         | 1.67             |
| Central | 204123                           | 2011      | 1400500         | 3.6                          | 496500                      | 30300                         | 1.74             |
| Central | 198471                           | 2012      | 1415950         | 3.6                          | 529333                      | 31517                         | 1.8              |
| Central | 175418                           | 2013      | 1592543         | 3                            | 593500                      | 32133                         | 1.82             |

Figure 3: The structure of the final dataset

Figure 3 represents the structure of my final dataset. The figure only shows the data for the Central region but the data for the other four regions are within the same dataset following the same structure.

## 2.3 Description of Variables

The response variable ( $Y$ ) is the number of crimes that occurred in that year. This includes different types of crimes such as robbery, burglary, fraud, violence, etc. The predictor variables ( $x_1, \dots, x_6$ ) are the factors that could influence the crime count.  $x_1$  is the specific "Year".  $x_2$  is the "population" of the five different regions within London.  $x_3$  is the "jobseeker's claimants" representing the percentage of people who claimed the unemployment benefit.  $x_4$  is the "median house price", which is the average house price within that region.  $x_5$  is the "average income" showing the average income for each individual within the working age. Lastly,  $x_6$  represents the "job density". This value shows the number of jobs available for every resident aged 16-64. *Note that "year" is used solely for producing the graphs and plots, so it is not really treated as a predictor.*

To improve interpretability, certain variables were rescaled. For example, the population values were originally in absolute counts (e.g. 1,400,000). These were then scaled to represent millions (e.g., 1.3) by dividing the original values by 1 million. The transformation allows the coefficients to be interpreted per 1 million increase within the population rather than each individual person. This makes it more meaningful and easier to interpret within the socio-economic context. Also,

the median house price and average income variables are scaled in the same way but to represent thousands.

### 3 Research Objectives

My analysis and study aims to provide insightful answers to the following questions:

- 1) How has crime in London changed over time (2001–2013) across the 5 different regions?
  - a) Are there any specific years with significant increases or decreases in crime rates across all regions?
  - b) Are there any potential reasons for the increase and decrease in part a)?
- 2) Which socio-economic variables affect the number of crimes?
  - a) What is the relationship between the socio-economic variables and the crime rate?
  - b) Which socio-economic variables are statistically significant predictors of crime?
- 3) Are there significant differences in the crime rate between the various regions after considering the socio-economic variables?
- 4) Did the 2008 financial crisis lead to vast changes in crime rates across London?
  - a) If so, were the impacts greater in some regions compared to others?
  - b) Are higher levels of jobseeker's claimants associated with higher crime rates during and after the financial crisis?
- 5) Do interactions between socio-economic variables (e.g. Job Density and Average Income) give a better insight into crime patterns compared to individual variables being analysed alone?

### 4 Methods

#### 4.1 Descriptive Analysis

In this section, I will talk through a few of the descriptive analysis methods computed to identify the general patterns and relationships within my data. This analysis presents a rough visualisation of the relationships, but it does not delve deeper into those trends.

##### **Choropleth Map**

*Note that 'City of London' is not a borough [Wikipedia contributors, 2025], so it is not included within this analysis and the blank space within the choropleth map is for this area.*

From figure (4), we can identify that there is a high number of crimes within the central boroughs

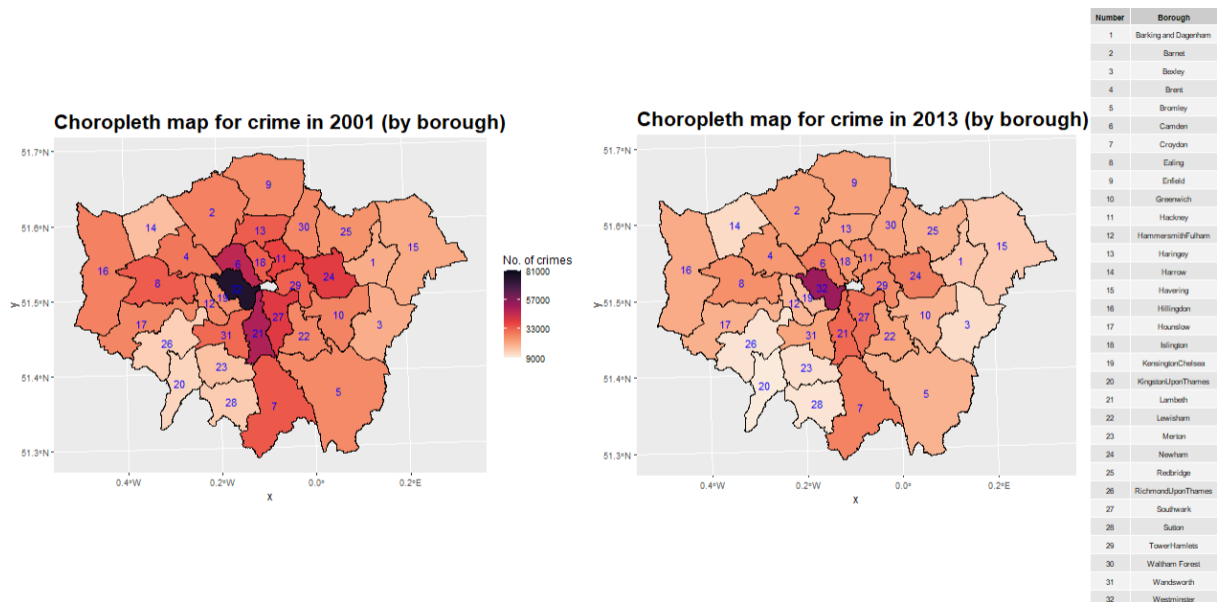


Figure 4: Choropleth map showing number of crimes in 2001 and 2013 (London boroughs)

of London and within all five sub-regions for 2001. However, if we compare it to 2013, we see that the number of crimes falls for most boroughs, if not all. Firstly, looking at the choropleth map for 2001, there are a few boroughs with high numbers of crime, which may inflate the crime count within that specific sub-region. These maps are used to identify hotspots (areas where the measuring factor is remarkably higher than other areas). In figure (4), it is easily noticeable that Westminster has an astonishingly high number of crimes (roughly  $> 70000$ ) compared to all other 31 boroughs. One of the main reasons for this is due to the 'west end'. The 'west end' is an area that is west of the City of London and within the boroughs of Camden and Westminster. It consists of many of London's tourist attractions, large expensive shops, government infrastructure, and mass amounts of entertainment venues. Areas within the borough of Westminster, like Oxford Street (Europe's busiest shopping street), attract large numbers of London residents and tourists, accumulating daily numbers of around 300,000 visitors. This is mainly due to the 300+ shops and the vibrant atmosphere of the street. However, alongside Westminster, there are two more regions that are standing out. Camden and Lambeth have a number of crimes slightly greater than around 45,000. Camden is another popular area for tourists because of the variety of markets, culture, and music venues alongside the British Museum and London Zoo. As a result of this, Camden has one of the highest theft crime rates within London [Bulbul, 2025]. The borough of Lambeth attracts many tourists because of the London Eye and Westminster Bridge (the bridge connecting Westminster and Lambeth).

Looking at the choropleth map for 2013, we see that the number of crimes has fallen across all boroughs. The number of crimes in Westminster has fallen to around 50,000. Similarly, for

Camden and Lambeth, the number of crimes for each borough has fallen to around 30,000. A possible reason for this could be the increase in funding for measures to tackle crime, such as extra police funding or more money to invest in improvements for security and safety. In recent times, between 2001 and 2024, Westminster had the highest crime rate within London, and theft was most common [Bulbul, 2025]. Crimes have been increasing non-stop since September 2024, so it is evident that since the fall between 2001 and 2013, there has been a rise from 2014 till the current day. Possible reasons for these rises are the increase in world-wide population leading to more potential tourists visiting these attractions within Westminster. Also, having a street of 300+ expensive and branded shops opens doors for some individuals to commit crime easily in the form of shoplifting.

From these choropleth maps, we can only identify the distribution of crime among the various boroughs within London. Apart from that, we cannot see the variations in the crime count between the boroughs as the years go by. The only way we can see the trends and variations is through creating choropleth maps for each year from 2001 to 2013, but this method is inefficient. Instead, we can look at the trends of sub-regions through line graphs.

### Crime trends over the years

For question 1, 'How has crime in London changed over time (2001–2013) across the 5 different regions?', a plot of the crime count over the years comparing all regions was created.

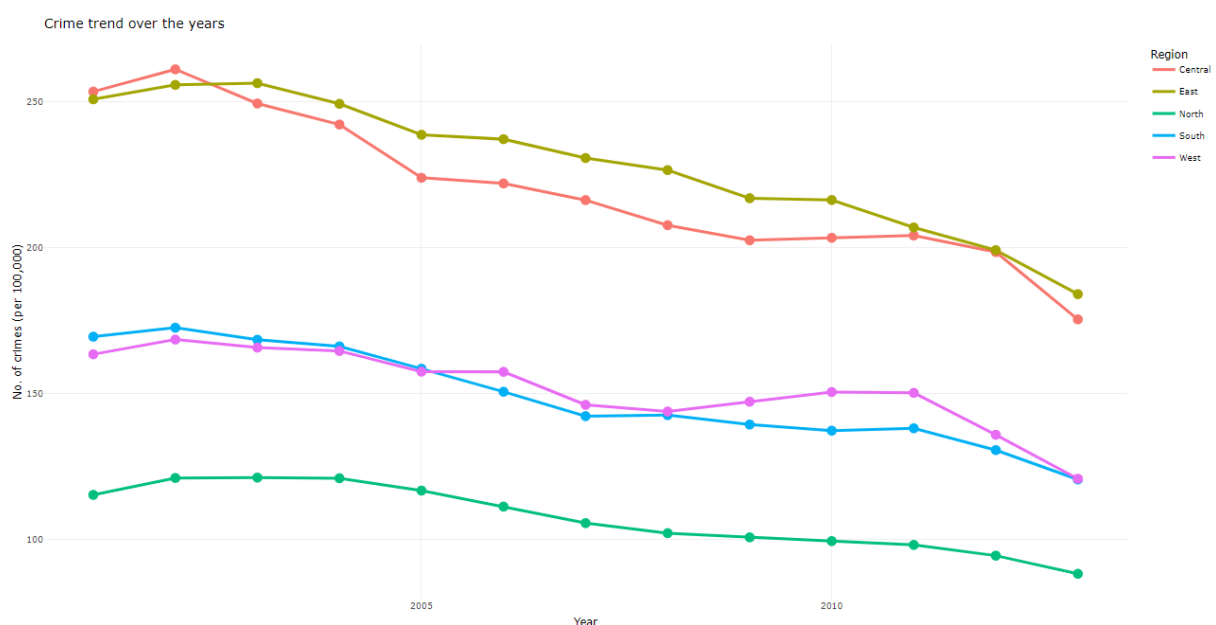


Figure 5: Crime trends over the years by region

From figure 5, we can see that the Central and West regions have the highest number of crimes throughout the 13 years, and the North region of London has the lowest number of crimes out

of all five regions. The overall trend for all regions from 2001 to 2013 is that the number of crimes start higher in 2001 and fall by the time they get to 2013. However, for some regions, there are fluctuations in the crime count. For example, for the Central region, between 2004 and 2005 there was a sharp decline in the crime count. A possible reason for this could be the extra police presence because of the 2005 terrorist attacks and the fall in the number of people visiting Central London after this tragic event. When we look at the fall in crime for the North region, the smooth decline suggests that there were no major events or incidents to cause sharp increases or declines within the trend line. For the West region, there is a period between 2008 and 2011 where the crime count increases. One reason for this could be the 2008 financial crisis, leading to mass amounts of unemployment within that region and providing more incentive to engage in criminal activity. The impact of this historic event will be talked about later on in this study. Therefore, this analysis of the crime trends across the years also answers question 1a of 'Are there any specific years with significant increases or decreases in crime rates across all regions?' and answers question 1b of 'Are there any potential reasons for the increase and decrease in part a?'.

## 4.2 Poisson Regression

### What is Poisson regression?

Poisson regression is a regression analysis method used for count and rate data. Count data models have dependent variables that are counts  $\{0,1,2,3,\dots\}$ . So the response variable ( $Y$ ) only takes non-negative integers. As suggested in [Roback and Legler, 2021], it may be argued that due to the trend of the data, using linear regression may be applicable to model the Poisson parameter. However, a linear model may not be effective for Poisson data as the line is going to sometimes provide negative  $Y$  values at given  $x$  values when  $Y$  can only take values in  $(0, \infty)$ . This does not make sense in the situation of crime analysis. For example, if linear regression was used to predict the number of crimes committed in a year, the results from linear regression may present a negative number of crimes, which does not logically make sense. But, we need to identify a few properties that may need to be considered when having a response variable that is count.

- The response is count, meaning that it cannot take negative values (this is discussed in detail later on).
- The model may be linear in  $x$  but it may also have some curvature.

- As  $x$  increases, the model may also appear to have data points more spread (increased variance).

So, this is why Poisson regression is a more frequently used method for response variables that are counts. We identify that Poisson regression models the linear relationship between a **response variable**, which is count, and various **predictor variables** that can be either numerical or categorical variables. However, one thing to consider is that the response variable is transformed using the natural log. The reason for this is because the response variable cannot be negative and linear regression can predict negative values for this. So, in order to restrict the response from being negative, the log is taken. Therefore,  $\log(Y)$  can now take any values in  $(-\infty, \infty)$ . This means the linear regression model can produce any value as it will be exponentiated to keep the count positive. Also, by doing this, the skewness of the data can be reduced and the data can be aligned better with the distribution.

### Poisson Distribution

The response variable is required to be Poisson distributed. Poisson distribution models the probability of a single event occurring a certain number of times within a fixed time interval. But two key points need to be considered:

- 1) The events have to occur independently (meaning the occurrence of one event does not affect the chance of the other event occurring).
- 2) The events occur at a constant average rate, which is given by  $\lambda$ .

The probability mass function (PMF) of  $Y$  distributed as Poisson with parameter  $\lambda$  is given by:

$$P(Y = y) = \frac{\lambda^y e^{-\lambda}}{y!} \quad (1)$$

where  $y = 0, 1, 2, 3, \dots$  and  $\lambda > 0$ .

Here,

- ' $Y$ ' is the number of events (in our case, it is the number of crimes in a region).
- ' $y$ ' is the observed value (meaning the actual data point from the dataset).
- ' $\lambda$ ' is the mean of  $Y$  (known as the rate).

A key property of this distribution is that the mean is equal to the variance (mean = variance =  $\lambda$ ).

### Where does the Poisson distribution arise from?

In fact, it is just a special case of the binomial distribution, where there is a large number of trials and small probability of success [Chamberlain, 2016]. As we know the average number of successes (rate of success) for a given time frame, but not the number of trials (n) of success probability (p), we can define the success rate per day to be,  $\lambda$ . This is equal to np because if the chance of success per day is p and there are n trials run per day, then the success rate ( $\lambda$ ) is:

$$\lambda = np \Rightarrow p = \frac{\lambda}{n}.$$

### Model equation

The Poisson regression model assumes that the count data ( $Y$ ) follows a Poisson distribution,  $Y_i \sim \text{Poisson}(\lambda_i)$ . The expected count is linked to the predictor variables using the log link function. Therefore, the Poisson regression model equation is:

$$\log(\lambda_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} \quad (2)$$

where,

- ' $Y_i$ ' is the number of events (e.g., number of crimes in a region).
- ' $\lambda_i$ ' is the expected value (mean) of the count variable (which is also equivalent to the variance).
- The log-link is included to make sure that  $\lambda_i$  is greater than zero and therefore the model is now log-linear.

### Assumptions for Poisson regression

There are key assumptions that need to be considered to ensure that the results obtained from the model are valid and reliable. Some assumptions are obtained from [Roback and Legler, 2021].

- **Response is assumed to be Poisson** - The response variable is a count per unit of time or space, described by a Poisson distribution.
- **Independence** - The observations must be independent of one another. This is to avoid 'false positives' (where a result is classified incorrectly as being true/positive when it actually is not). This cannot be validated statistically but can be verified through how the data was collected.
- **Mean = Variance** - For given  $x$ 's, by definition, the mean of a Poisson random variable must be equal to its variance. In other words,  $E[Y] = \text{var}[Y] = \lambda$ , so there is no over or under dispersion.



- **Linearity** - The log of the mean rate,  $\log(\lambda)$ , or  $\log(\text{count})$  must be a linear function of  $x$ .  
This assumption can be validated through looking at residual plots.

### Checking whether the data meets the assumptions

We can check whether the data meets the assumptions for Poisson regression.

To check whether the **response is assumed to be Poisson distributed**, I plotted the histogram for the Crime response variable ( $Y$ ) and the Poisson distribution curve (based on the mean of the data) to check whether they match. If they follow the same shape, then the response variable is assumed to be Poisson distributed.

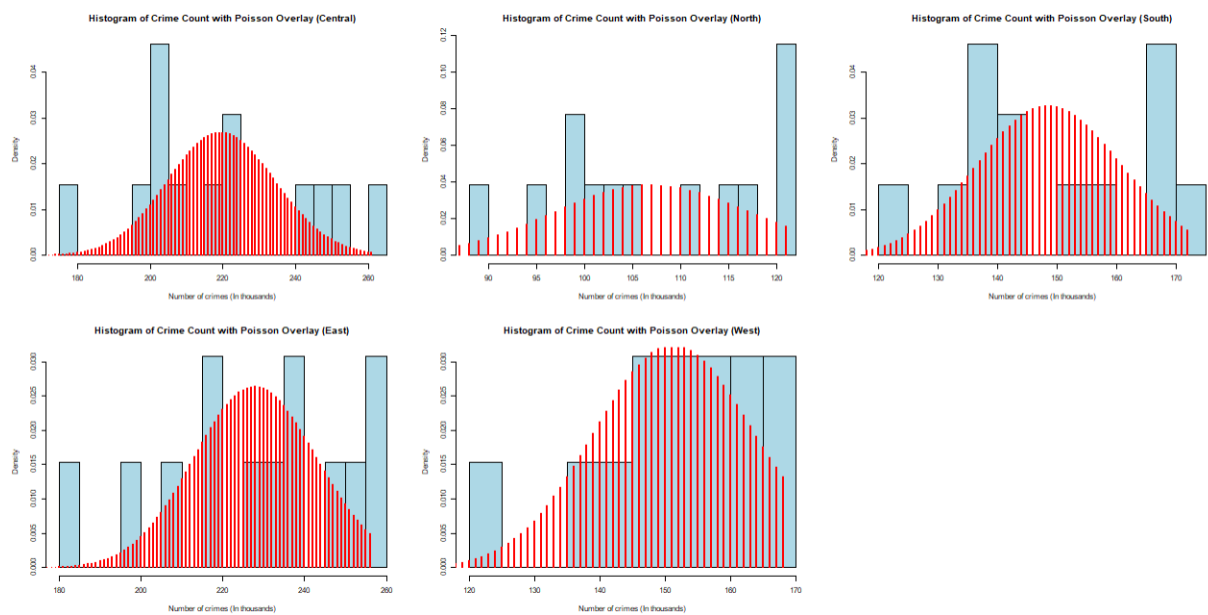


Figure 6: Histogram for Crime with Poisson overlay (By region)

From figure 6, it can be identified that some of the regions have a response variable that is Poisson distributed but there is skewness involved. For example, in the histogram for the West of London, the histogram is highly left-skewed, but the Poisson distribution curve is also slightly left-skewed, meaning that we can assume that the response for that region is Poisson distributed. Similarly, for the East of London, it can be seen that the Poisson distribution curve is following a similar shape to the histogram. This leads us to assume that the response is Poisson distributed. However, this is not the case for all regions. For some regions, the Poisson curve is not a similar shape to the corresponding histogram, meaning that the response may not be Poisson distributed and could hinder the validity of the regression.

This assumption of assuming the response is Poisson distributed can also be verified through looking at the Q-Q plots for the various regions from figure 7. Q-Q (Quantile-Quantile) plots are

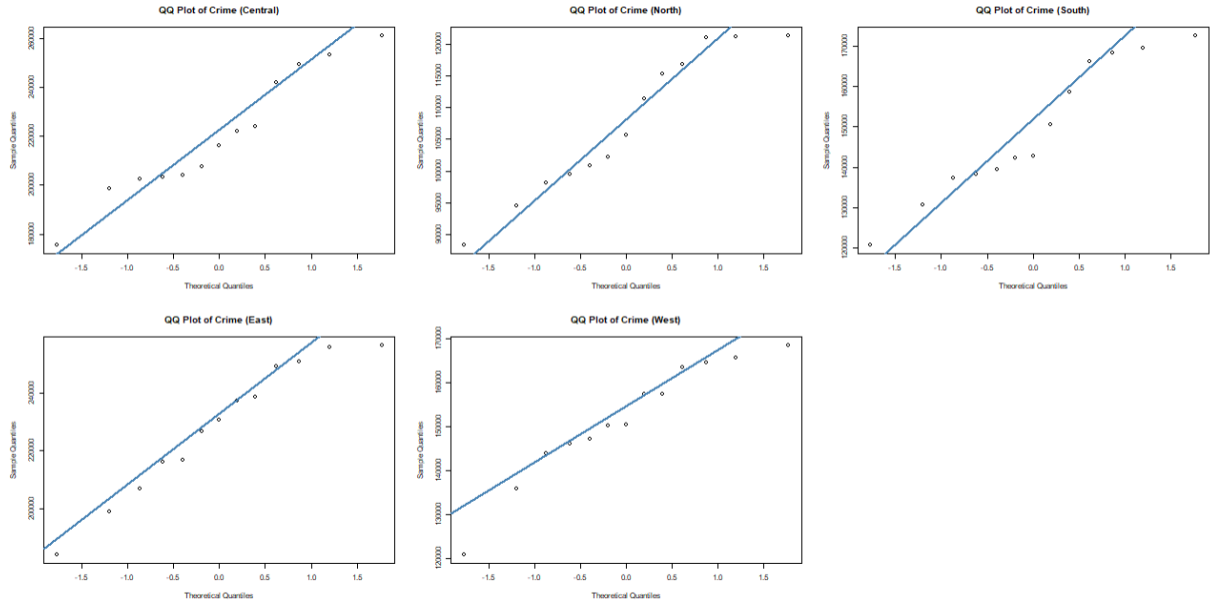


Figure 7: Q-Q Plots for Crime (By region)

used to visualise whether the data follows a specific distribution. In our case, if the response is Poisson distributed, then the points should roughly fall on the diagonal line [García-Portugués, 2025]. From the Q-Q plots, it can be seen that for the West and East of London, the points roughly fall on the diagonal line. However, there are a few points that deviate from the line near the tails. This suggests that there is some overdispersion involved and these deviations at the tails may influence the skewness that we saw in the histograms for these regions. If we look at the Q-Q plots for the other three regions, there are a few points that 'dip' and some 'arch'. This suggests that there is some skewness involved within the data.

Another assumption is that the **observations are independent of one another**. At first glance, we would assume that the observations of the crime number are completely independent of each other as they are not clustered or providing similar values to the rest of the observations. However, as we are looking at the crime number for each year, from 2001 to 2013, there is a possibility that the amount of crimes committed in the previous years may affect the number of crimes committed in the current year. For example, if crime was extremely high one year, then the following year it could fall by a large number due to certain measures being put into place to lower crime. So, there may be dependence over a period of time. One way to check this is through creating autocorrelation plots from the time-series models for the different regions. Autocorrelation measures the relationship between a time series model and a lagged version. It represents how similar the data points are to each other at different time lags [Kis, 2024]. The autocorrelation plot shows the correlation of the time series models at different lag values, allowing us to visualise trends and whether there is dependence during the series. Looking at the

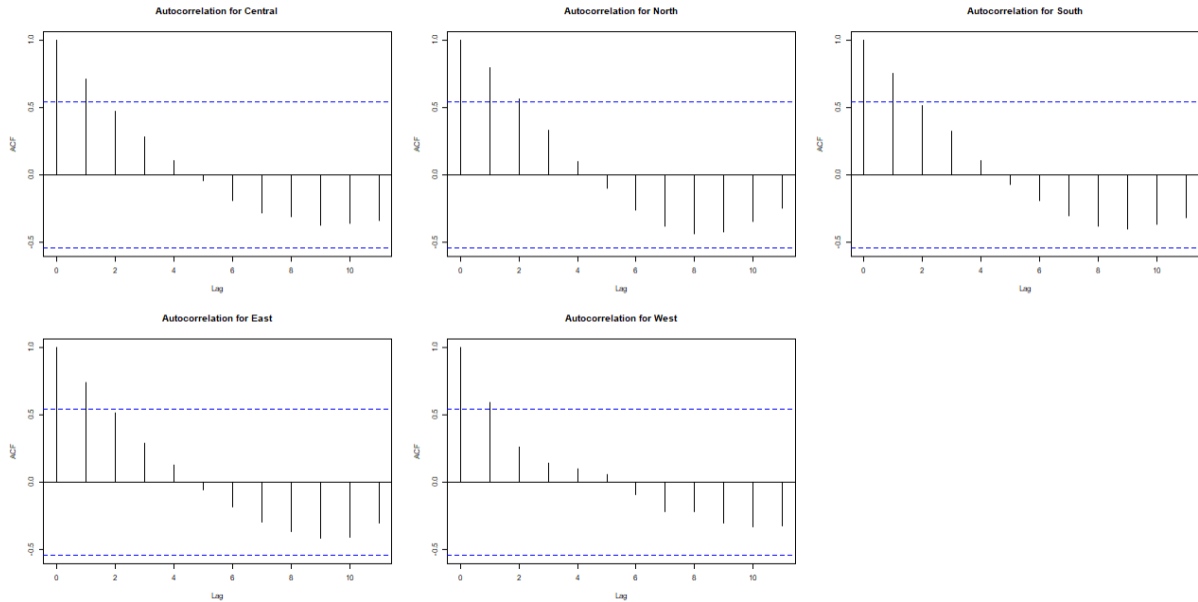


Figure 8: Autocorrelation Plot for Crime (By region)

autocorrelation plots from 8, the bar represents the value of the correlation at that lag value. One thing to note is that for lag 0, the correlation is most likely always going to be 1 because we are comparing 'year  $t$ ' to 'year  $t$ ', which should always have a correlation of 1 as we are comparing each term (in our case 'year') to itself. The blue boundaries represent the 95% confidence interval. The values outside these boundaries suggest autocorrelation at that specific lag. Now, looking at the autocorrelation for the different regions in our plot, we see that for all of the plots the values stay within the boundaries, except from lag 1 and 2 for specific regions. This suggests that there is some autocorrelation involved when comparing the current year to the previous year or to two prior years.

Another test computed to identify whether there was autocorrelation within the residuals is the 'Darwin-Watson test'. This test was used to see if there is first-order correlation (comparing current year to previous year). When this test was computed, we resulted in a p-value of 0 for ALL regions. This instantly meant that there was autocorrelation involved for lag 1.

*Note: these tests are used for identifying autocorrelation. We can still assume that the observations are independently obtained.*

Next, to check the assumption of assuming that **mean being equal to variance**, we can simply compute the mean and variance for the response variable for each region. If the variance is equal to the mean, then this assumption is valid. However, if the variance is less than or greater than the mean then there is underdispersion or overdispersion respectively. When computing the mean and variance for response variable of the number of crimes, which can be viewed in

| Region  | Mean     | Variance  |
|---------|----------|-----------|
| Central | 219959.6 | 634893540 |
| North   | 107399.8 | 123538041 |
| South   | 148997.0 | 276356312 |
| East    | 228315.6 | 517415336 |
| West    | 151708.3 | 180653772 |

Table 1: Mean and Variance of Crime response variable

table 1, it was quite evident that the variance was extremely greater than the mean for each corresponding region. This gives insight into the fact that there is overdispersion involved and there may be better regression models that we could fit compared to Poisson regression.

The last assumption we test is linearity. One way we test this is through plotting the log of the mean crime count against the different predictor variables. Figure 6 shows the plot we use to identify whether this assumption is met. We can see that most of the plots show a linear relationship between the log of the mean count and the five predictor variables over the years. Therefore, we can say that we believe this assumption of linearity is met.



Figure 9: Log of mean count plot for Crime (By region over time)

We can further validate this assumption through looking at the residual plots from 10.

Residual plots present the fitted values against the predicted values. It allows us to assess linearity by presenting the residuals to be randomly scattered (indicating a linear relationship) or to be following a certain pattern (indicating a non-linear relationship). From the residual plots, if the points show no pattern and are randomly scattered, then we assume that there is a linear

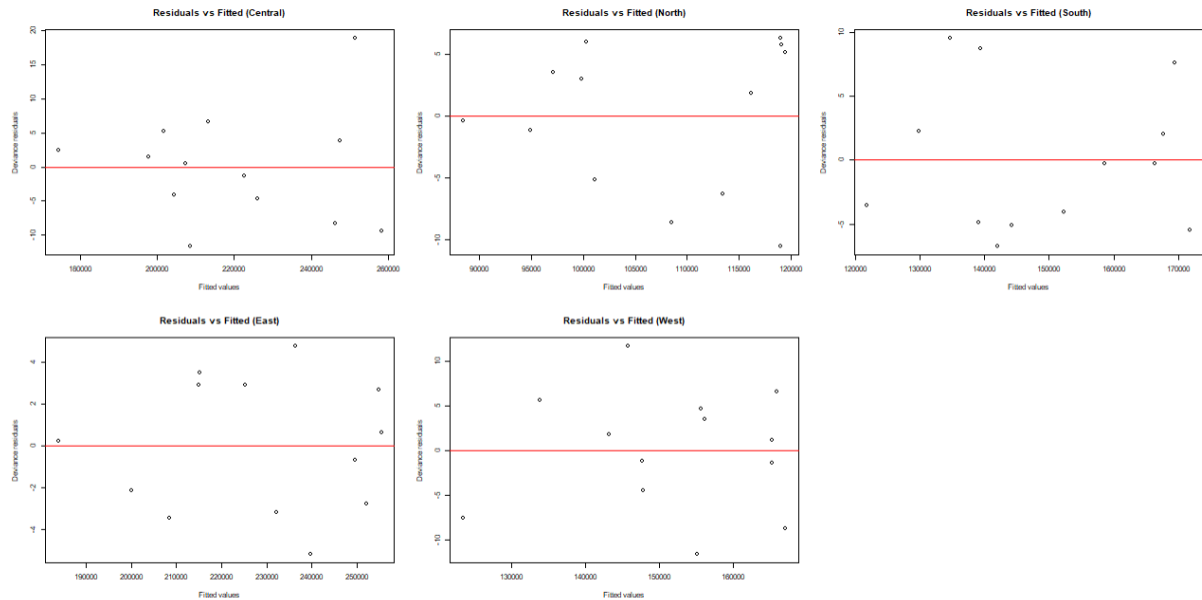


Figure 10: Residual plot for Crime (By region)

relationship. We see that there is no particular pattern within the residual plots, so we can assume that the linearity assumption is met. This assumption is particularly important for Poisson regression as we need to identify whether the log-link function has a linear relationship with the predictor variables.

From testing all of these assumptions, we were clearly able to identify that some assumptions were met and some were not. However, in some cases, the assumptions were met for some regions and were not met for others. For example, when looking at the assumption for the response to be Poisson distributed through the histograms, we were able to identify that for certain regions like West and East of London, the Poisson distribution curve followed a similar shape to the corresponding histograms allowing us to assume that this assumption is met for these two regions. However, for the other three regions, the Poisson distribution curve did not follow the shape of the histogram leading us to assume that it may not meet the assumption. But, by just looking at this one method, we cannot be certain. Therefore, through the Q-Q plots we are able to identify that there was some skewness involved for the three regions potentially swaying us further away from assuming that the response is Poisson distributed for Central, North, and South of London. For the independence assumption, we were able to use autocorrelation plots to identify that there was correlation between the current year and the previous one or two years for all five regions. But, when we compare the current year to the previous three, four, or all the way up to twelve years, we see that the correlation becomes really small over time and eventually becomes negative for all regions. This suggests that there is only correlation for the current year and the previous one or two years. Regarding the assumption for assuming mean and variance

to be equivalent, it was quite evident that there was overdispersion for all five regions. Lastly, we were able to assume that the linearity relationship was met between the log mean count and the various predictor variables. This was supported by the plot showing the linear relationships between the log mean count and the five predictor variables throughout the years for all regions.

So what can we take from this? One key thing to take away from these results is that a few assumptions were met and a few were not. While we saw the assumption of Poisson distributed response (for some regions) and linearity being met, overdispersion and dependence between certain years was still present. Thus, especially due to the overdispersion being present, it would be better to use models that can handle the overdispersion and the violated assumptions, such as Negative Binomial regression. The assumptions being violated will not mean that the Poisson regression models do not work. They will provide results when they are computed. However, the accuracy and validity of the model will be affected due to the assumptions being violated.

### 4.3 Negative Binomial Regression

#### What is Negative Binomial regression?

Negative Binomial regression is another method used to model count data, similar to Poisson regression. However, the key difference between both methods is that Negative Binomial does not require the mean and variance to be equal. Negative Binomial regression addresses overdispersion in data, where variance is greater than the mean. So, this implies that if the data does not satisfy the Poisson regression assumption of equidispersion and the variance is greater than the mean, then Negative Binomial is a better method to model with. It is also a more flexible alternative to Poisson regression. Negative Binomial regression arises from its distribution. Negative Binomial distribution is used to determine the number of trials needed in order to achieve a target number of successes.

The following formula suggested in [Zwilling, 2013] is for the Negative Binomial distribution:

$$P(Y = y) = \frac{\Gamma(y + \frac{1}{\alpha})}{\Gamma(y + 1)\Gamma(\frac{1}{\alpha})} \left( \frac{1}{1 + \alpha\mu} \right)^{\frac{1}{\alpha}} \left( \frac{\alpha\mu}{1 + \alpha\mu} \right)^y. \quad (3)$$

Here,  $\mu > 0$  is the mean of  $Y$  and  $\alpha > 0$  is the heterogeneity parameter. This parameter is a case where the regression coefficients of the model differ across observations within the data, due to various factors. Hilbe suggests it is understood that the Negative Binomial distribution is a Poisson-gamma distribution mixture model.

In line with the findings of [Pesta, 2020], the Poisson regression model could be generalised through the introduction of an 'unobserved heterogeneity term' for the  $i_{th}$  observation. This term describes the existence of unmeasured differences between samples that are being studied, which affect the outcome but are not captured by the observed variables. Therefore, the individuals are assumed to differ, which is not fully accounted for by the covariates. This assumption is represented through the formulation:

$$\mathbb{E}[Y_i|x_i, \tau_i] = \mu_i \tau_i = e^{x_i^T \beta + \epsilon_i},$$

where  $\tau_i = e^{\epsilon_i}$  is the heterogeneity term and is independent of the  $x_i$ 's. The distribution of  $Y_i$  conditioned on  $x_i$  is:

$$f(y_i|x_i, \tau_i) = \frac{e^{-\mu_i \tau_i} (\mu_i \tau_i)^{y_i}}{y_i!}, \quad y_i = 0, 1, 2, \dots$$

Here,  $\tau_i$  is Poisson and the mean and variance are both  $\mu_i \tau_i$  (which can be identified by comparing this equation to the general Poisson distribution formula). Now, we let the probability density function of  $\tau_i$  be  $g(\tau_i)$ . Using this, we can obtain the distribution of  $f(y_i|x_i)$  by integrating the equation we had for  $f(y_i|x_i, \tau_i)$  with respect to  $\tau_i$ :

$$f(y_i|x_i) = \int_0^\infty f(y_i|x_i, \tau_i) g(\tau_i) d\tau_i.$$

The solution to this integral exists when  $\tau_i$  follows gamma distribution. Thus, the solution from this integral is the Negative Binomial distribution. When the model has a constant, we assume  $\mathbb{E}_{\tau_i} = 1$ , which is used to find the mean of the distribution. Therefore, we assume  $\tau_i$  follows a gamma  $(\theta, \theta)$ . Recall, for a general  $X \sim \text{Gamma}(\alpha, \lambda)$ , the expectation and variance of  $X$  are given by the following:  $\mathbb{E}(X) = \frac{\alpha}{\lambda}$  and  $\text{Var}(X) = \frac{\alpha}{\lambda^2}$ . Relating this back to  $\tau_i$ ,  $\mathbb{E}(\tau_i) = 1$  and  $\text{Var}(\tau_i) = \frac{1}{\theta}$ . Thus, the probability density function of the gamma distribution can be written as:

$$g(\tau_i) = \frac{\theta^\theta}{\Gamma(\theta)} \tau_i^{\theta-1} e^{-\theta \tau_i}.$$

In this equation,  $\Gamma(\theta) = \int_0^\infty z^{\theta-1} e^{-z} dz$  is the gamma function and  $\theta$  is positive. Then, the probability density of  $(Y_i|x_i)$  is:

$$f(y_i|x_i) = \frac{\Gamma(y_i + \theta)}{y_i! \Gamma(\theta)} \left( \frac{\theta}{\theta + \mu_i} \right)^\theta \left( \frac{\mu_i}{\theta + \mu_i} \right)^{y_i}, \quad y_i = 0, 1, 2, \dots$$

Therefore, to obtain the Negative Binomial distribution, we can make the substitution of  $\alpha = \frac{1}{\theta}$ , where  $\alpha > 0$  and we get:

$$f(y_i|x_i) = \frac{\Gamma(y_i + \alpha^{-1})}{y_i! \Gamma(\alpha^{-1})} \left( \frac{\alpha^{-1}}{\alpha^{-1} + \mu_i} \right)^{\alpha^{-1}} \left( \frac{\mu_i}{\alpha^{-1} + \mu_i} \right)^{y_i}, \quad y_i = 0, 1, 2, \dots$$

Therefore, we have completed the derivation of the Negative Binomial distribution. It can be seen as a gamma mixture of Poisson random variables, which we mentioned before. The conditional mean is

$$\mathbb{E}[Y_i|x_i] = e^{x_i^T \beta}$$

and conditional variance

$$Var[Y_i|x_i] = \mu_i \left( 1 + \frac{\mu_i}{\theta} \right) = \mu_i(1 + \alpha\mu_i) = \mu_i + \alpha\mu_i^2$$

The conditional variance is greater than the conditional mean and here is where the term 'overdispersion' arises from. The overdispersion is a result of the unobserved heterogeneity term. The Negative Binomial model with the variance mentioned above is referred to as 'NB2'. *Note that in the calculations computed above for the Negative Binomial distribution,  $\alpha$  is equivalent to  $\frac{1}{\theta}$ .*

### Model Equation

The Negative Binomial regression considers overdispersion. So, it is an extension of the Poisson regression as it introduces a dispersion parameter ( $\theta$ ). The count response variable ( $Y_i$ ) has a Negative Binomial distribution,  $Y_i \sim NB(\mu_i, \theta)$ . Thus, the Negative Binomial regression model equation is:

$$\log(\mu_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} \quad (4)$$

where,

- ' $\mu_i$ ' is the mean of the distribution (Negative Binomial distribution in this case).
- ' $\theta$ ' is the overdispersion parameter for the variance being greater than the mean.
- Again, log-link is included to make sure that  $\mu_i > 0$  and therefore provides a log-linear model.

We see that the equations for Poisson regression and Negative Binomial regression are the same. However, the key difference is the dispersion parameter ( $\theta$ ) representing the variance no longer being equal to the mean. The variance is now greater than the mean,  $Var(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$ , where  $\theta$  measures the overdispersion (the larger the  $\theta$ , the less overdispersion).



## Assumptions for Negative Binomial regression

Just like the assumptions for Poisson regression, there are certain assumptions that need to be met for the Negative Binomial models.. These assumptions are building on from [Rahul, 2023]. As Poisson regression is a special case of Negative Binomial regression, where the dispersion parameter is zero, there are some assumptions that still hold for Negative Binomial regression. For example, the assumption of assuming that the observations are independent of each other still holds here. So we don't need to test this. The other assumption of linearity still holds here, and we saw that this assumption had been met, so we don't need to test this again either. Now, the next few assumptions are specifically for Negative Binomial regression:

- **No Multicollinearity** - The predictor variables should not have strong correlation with the other predictors. This is because we won't be able to identify the individual effect of each predictor variable on the response variable.
- **Overdispersion** - The model is used to overcome the issue of overdispersion. Therefore, the variance of the response must be greater than it's mean for us to use this model.

## Checking whether our data meets the assumptions

To check whether the assumption of **no multicollinearity** is met, a few methods were used. The first method was checking the correlation between the predictor variables through computing the correlation matrix and the correlation plots.

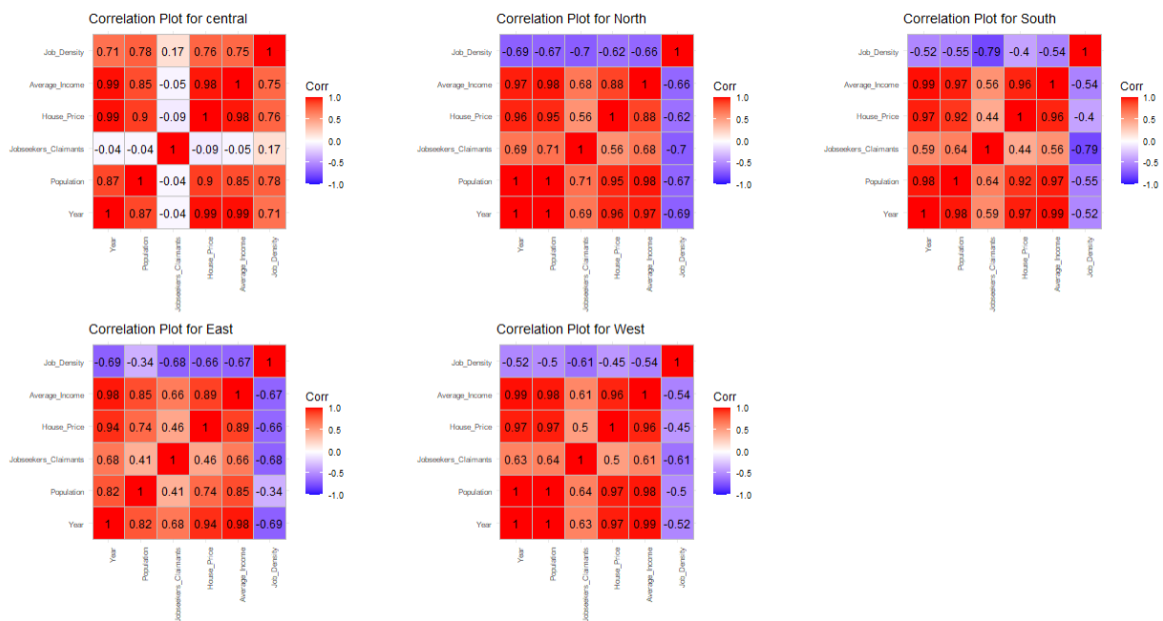


Figure 11: Correlation Plot of the predictor variables (By region)

From figure 11, it is clearly visible that there is a correlation between some predictors. However, there is an interesting result to notice in the correlation plot for Central. The jobseeker's

claimants predictor has almost no correlation with any of the other five predictors. This suggests that there is very little association between the jobseeker's claimants and the other predictors. Whereas, if we look at the same predictor for the rest of the regions, it has a positive correlation with the other predictors, except for the Job density predictor.

The second method is calculating the **Variance Inflation Factor** [Bobbitt, 2022]. The Variance Inflation Factor (VIF) measures the strength of correlation between the various predictor variables involved within the model. Generally, a VIF value of 1 tells us that there is no correlation between the specific predictor and the other predictors in the model. A VIF value between 1 and 5 means that there is slight correlation between the specific predictor and the other predictors. However, a VIF value greater than 5 shows that there is a strong correlation (multicollinearity).

| <b>Region</b> | <b>Population</b> | <b>jobseeker's<br/>Claimants</b> | <b>Median<br/>house<br/>price</b> | <b>Average<br/>income</b> | <b>Job<br/>density</b> |
|---------------|-------------------|----------------------------------|-----------------------------------|---------------------------|------------------------|
| Central       | 8.44              | 1.18                             | 51.69                             | 34.59                     | 3.24                   |
| North         | 118.54            | 4.82                             | 26.47                             | 40.20                     | 2.67                   |
| South         | 21.26             | 3.95                             | 19.09                             | 43.73                     | 3.55                   |
| East          | 5.95              | 2.93                             | 6.45                              | 16.47                     | 3.56                   |
| West          | 94.43             | 4.50                             | 29.02                             | 45.41                     | 2.41                   |

Table 2: Results from undertaking VIF using Negative Binomial models

From table 2, we clearly see that the values for certain areas and predictors are greater than 5. For example, if we look at the average income predictor for all the areas, we see that the values are large, leading us to assume that there is multicollinearity involved. Also, other predictor variables such as median house price have large VIF values, so we can definitely see that there is multicollinearity.

One way to resolve this problem is by removing the predictor variable (or more than one variables), that are highly correlated. This is acceptable to do because this variable that is being removed does not have much of an impact on the model by itself. So, when removing the average income predictor, we obtained the following results:

From table 3, it is obvious that all the values have fallen compared to what was obtained from table 2. This is mainly due to the predictor that was highly correlated being removed, leading to the values to fall for the remaining predictors. We can see that now most values are either

| Region  | Population | jobseeker's<br>Claimants | Median<br>house<br>price | Job<br>density |
|---------|------------|--------------------------|--------------------------|----------------|
| Central | 6.27       | 1.16                     | 5.88                     | 3.05           |
| North   | 17.61      | 3.43                     | 13.37                    | 2.33           |
| South   | 11.27      | 3.73                     | 8.24                     | 2.70           |
| East    | 2.78       | 2.17                     | 4.09                     | 3.29           |
| West    | 35.44      | 3.73                     | 28.20                    | 1.74           |

Table 3: Results from VIF using Negative Binomial models without 'Average income'

smaller than 5 or slightly greater than 5, with an exception of a few values. From these calculations, we are able to conclude that there is multicollinearity between the predictors but it does fall when these predictors with high multicollinearity are removed.

The other assumption we need to consider is **overdispersion**. For this assumption, we already tested it when we were checking the assumption for the mean and variance being equivalent in Poisson regression. When we calculated the mean and variance of the response for the different areas, the variance was greater than the mean. Also, various different tests were computed like the 'Cameron and Trivedi dispersion test' and these provided results in favour of overdispersion. Therefore, overdispersion was present within the data and this assumption is met.

Through undertaking various tests to identify whether the assumptions were met, we were able to see that all assumptions for Negative Binomial regression were met except for no multicollinearity. The assumptions for observations being independent and linearity were already met from Poisson regression. However, we were able to notice that through the correlation plots and VIF values, there was multicollinearity involved for certain predictors. Lastly, the assumption of overdispersion was obvious when we computed the variance and mean of the response variable.

## 4.4 Beta Regression

### What is beta regression?

Beta regression is used to model variables that are assumed to have values within the unit interval  $(0, 1)$ . We adapted the methodology described in [Cribari-Neto and Zeileis, 2010] to understand the components within Beta regression. Similar to the other regression methods mentioned in this paper, this method assumes the response variable to be beta-distributed. If the response

variable doesn't fall within the interval  $(0, 1)$  and takes on values within an interval  $(a, b)$ , where  $a < b$ , then we can use a technique called 'Min-Max Scaling' to standardise the values within the response variable. We use the following formula to standardise the values:

$$Y_{scaled} = \frac{Y - \min(Y)}{\max(Y) - \min(Y)}$$

where  $Y$  is the original response values,  $\min(Y)$  is the minimum value out of all the  $Y$  values and  $\max(Y)$  is the maximum value out of all  $Y$  values. Therefore, the formula is applied to each individual  $Y$ , which will then provide the scaled  $Y$  values within the  $(0, 1)$  interval.

The following formula is for the probability density function of the beta distribution:

$$f(y; p, q) = \frac{\Gamma(p+q)}{\Gamma(p)\Gamma(q)} y^{p-1} (1-y)^{q-1}, \quad 0 < y < 1. \quad (5)$$

Here,  $p, q > 0$  and as seen before,  $\Gamma(\cdot)$  is the gamma function. The mean and variance of  $y$  is given by the following equations:

$$\mathbb{E}(y) = \frac{p}{(p+q)}$$

and

$$var(y) = \frac{pq}{(p+q)^2(p+q+1)}.$$

Using the beta-distributed random variables for the response ( $y$ ), we want to identify the corresponding regression model. The original beta distribution from equation (5) is dependent on  $p$  and  $q$ . But similar to all the previous regression models mentioned, the regression is more useful when modelling the mean of the response. When defining the beta regression model, we normally require it to contain a dispersion (or precision) parameter, which describes the scatter of individual data around the mean. This parameterisation was proposed by [Ferrari and Cribari-Neto, 2004], where they adjusted the original beta density model by letting  $\mu = \frac{p}{(p+q)}$  and  $\phi = p + q$  and provided the following alternative parameterisation:

$$f(y; \mu, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma((1-\mu)\phi)} y^{\mu\phi-1} (1-y)^{(1-\mu)\phi-1}, \quad 0 < y < 1. \quad (6)$$

In this equation,  $\phi > 0$  and  $0 < \mu < 1$ . Using these new parameters, the response being beta distributed can be written as  $y \sim B(\mu, \phi)$ . The mean and variance become  $\mathbb{E}(y) = \mu$  and  $var(y) = \frac{\mu(1-\mu)}{(1+\phi)}$ . Given a fixed  $\mu$ , if the precision parameter ( $\phi$ ) is large, then the variance of  $y$  is small. In order to define the beta regression model, the response variable needs to be defined. We let  $y_1, \dots, y_n$  be the response variable and assume them to be beta distributed,  $y \sim B(\mu, \phi)$ , with  $i = 1, \dots, n$ . Then, the beta regression model can be formalised as

$$g(\mu_i) = x_i^T \beta = \eta_i. \quad (7)$$

Defining the various parameters within the equation,  $\beta = (\beta_1, \dots, \beta_k)^T$  is a  $k \times 1$  vector of the regression coefficients that need to be calculated. The vector  $x_i = (x_{i1}, \dots, x_{ik})^T$  is the vector of the predictor variables that are being used in the regression.  $\eta_i$  is the linear predictor term. So, the product of the  $\beta$  vector and the  $x_i$  vector ( $\eta_i = \beta_1 x_{i1} + \dots, \beta_k x_{ik}$ ). One aspect to note is that the first term  $x_{i1}$  tends to be 1 for all  $i$  values to confirm that the model has an intercept term.

Just like with the other regression methods within this paper, this method also uses a link function. The most commonly used link functions are: the logit,  $\eta = \log\left(\frac{\mu}{1-\mu}\right)$ ; complementary log-log link,  $\eta = \log(-\log(1 - \mu))$ ; and log-log link,  $\eta = -\log(-\log(\mu))$ . The log-likelihood function is

$$\ell(\beta, \phi) = \sum_{i=1}^n \log \Gamma(\phi) - \log \Gamma(\mu\phi) - \log \Gamma((1 - \mu_i)\phi) + (\mu_i\phi - 1)\log(y_i) + ((1 - \mu_i)\phi - 1)\log(1 - y_i). \quad (8)$$

Note that, when we obtained the beta regression model, we could re-arrange the formula to see that  $\mu_i = g^{-1}(x_i^T \beta)$ .

### Model Equation

The beta regression is used for rates that are bounded between 0 and 1. In our case, the crime rate is scaled between 0 and 1. The continuous response variable ( $Y_i$ ) has a Beta distribution,  $Y_i \sim \text{Beta}(\mu_i, \phi)$ . Therefore, the Beta regression model equation is:

$$\text{logit}(\mu_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} \quad (9)$$

where,

- ' $Y_i$ ' is the continuous variable in the range (0,1).
- ' $\mu_i$ ' is the mean of the distribution (Beta distribution in this case).
- The logit link function is included to transform the  $\mu$  values that are between 0 and 1 to values between  $-\infty$  and  $\infty$ .

The link function used in this equation could be the various ones mentioned earlier but the most common used is the 'logit function'. *Note that  $\text{logit}(\mu_i) = \log\left(\frac{\mu_i}{1-\mu_i}\right)$ .*

### Assumptions for Beta regression

There is a set of assumptions that need to met for Beta regression to identify whether the results would be valid and reliable. These assumptions are similar to what we have seen throughout this research for the other two methods, but now relating to beta distribution and the scaled response variable. The assumptions include:

- **Response is assumed to be Beta distributed** - The response variable (Y) is assumed to be following Beta distribution because Beta regression uses Beta distribution to model response variables between 0 and 1.
- **Response values are in (0,1)** - The values have to be within this interval because we are using beta distribution, which is defined on this interval.
- **No Multicollinearity** - The predictor variables should not have strong correlations with the other predictors due to the same reason as mentioned in the Negative Binomial assumptions.
- **Linearity in link function** - The link function of the mean of the response should be a linear combination of the predictors.

### Checking whether our data meets the assumptions

Model diagnostics are used to identify how well the model fits the data. If the model is poorly fit, then we consider making adjustments to the model and then re-fit it to the data to see whether improvements are made.

To check whether the **response is assumed to be Beta distributed**, I plotted the histogram of the scaled response variable (Y) and the Beta distribution to identify whether they follow a similar shape.

From figure 12, we can see that for some regions, the response variable follows the Beta distribution slightly. For example, we see that for South, East and West the beta curve follows the shape of the corresponding histograms. For these regions, the Beta curves peak at the same places as when the histograms peak. This suggests that we could assume the scaled crime rates are Beta distributed for these specific regions. However, if we look at the other two regions, we see that the beta curves do not peak when there is a peak in the histograms, leading us to assume

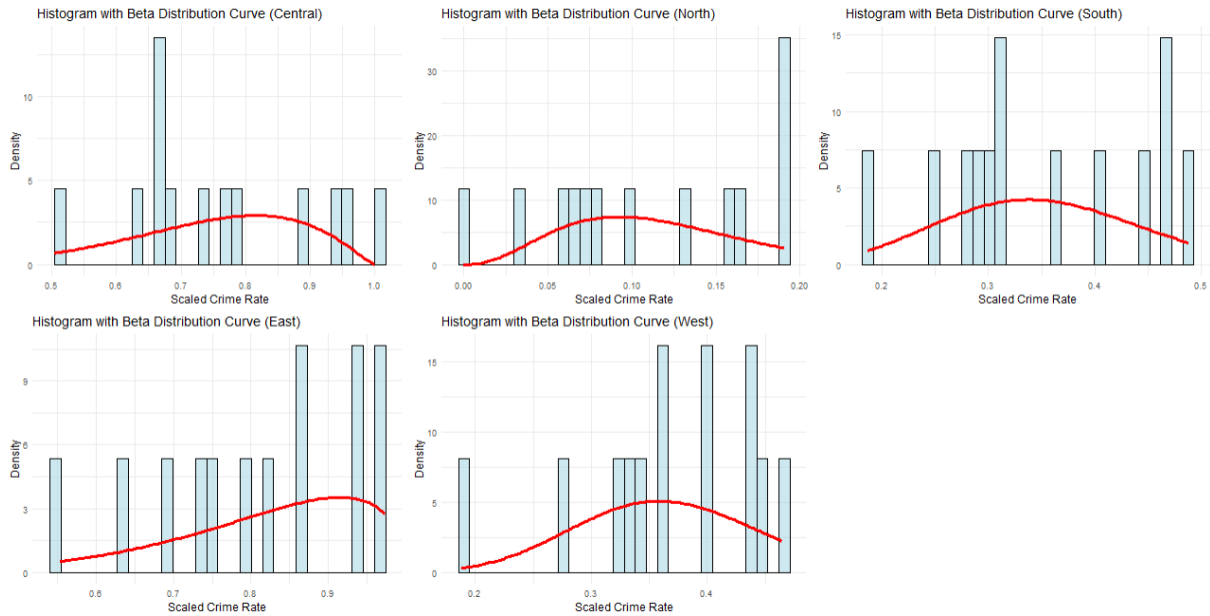


Figure 12: Histogram for scaled Crime with Beta overlay (By region)

that these two areas have a response variable that is not Beta distributed.

Next, the assumption of assuming the **response values are in the interval of (0,1)** can be tested through looking at the response variable after it has been scaled. We used the min-max scaling method to scale the original response variables to fall within the interval (0,1). When we looked at the values for the response variable, all the values were within this interval suggesting that this assumption has been met.

To identify whether the assumption of **no multicollinearity** has been met, we use the same method that was used in section 4.3 for checking the assumption for Negative Binomial regression (Variance Inflation Method). We are looking to avoid predictors that provide a VIF value greater than 5.

| Region  | Population | jobseeker's<br>Claimants | Median<br>house<br>price | Average<br>income | Job<br>density |
|---------|------------|--------------------------|--------------------------|-------------------|----------------|
| Central | 8.28       | 1.13                     | 50.64                    | 44.55             | 6.57           |
| North   | 11.95      | 2.51                     | 5.13                     | 9.32              | 2.55           |
| South   | 21.07      | 3.73                     | 18.69                    | 44.26             | 3.58           |
| East    | 6.53       | 3.44                     | 7.92                     | 19.58             | 2.81           |
| West    | 87.95      | 4.39                     | 26.60                    | 43.94             | 2.68           |

Table 4: Results from undertaking VIF using Beta models

From table 4, we can identify that there are specific predictors that provide large VIF values. If we look at the average income predictor, the values are clearly quite large compared to the other predictor. This suggests that there is multicollinearity and this predictor by itself is not significant as it relies on other predictors within the model. When we remove this predictor, we result in the following table:

| Region  | Population | jobseeker's<br>Claimants | Median<br>house<br>price | Job<br>density |
|---------|------------|--------------------------|--------------------------|----------------|
| Central | 4.41       | 1.10                     | 4.29                     | 3.44           |
| North   | 25663.07   | 15.02                    | 3.43                     | 25190.42       |
| South   | 10.87      | 3.47                     | 7.80                     | 2.66           |
| East    | 3.16       | 1.95                     | 3.78                     | 2.77           |
| West    | 33.09      | 3.64                     | 26.31                    | 1.83           |

Table 5: Results from VIF using Beta models without 'Average income'

From table 5, we once again see that the VIF values have fallen for all predictors and regions, except one region. For North, the values have instead risen by large amounts. This suggests that by removing the average income predictor, the multicollinearity has become worse. Therefore, we can conclude that this assumption is not met and there is multicollinearity involved, which may affect the validity of the results.

Lastly, we can assess the assumption of **assuming linearity in the link function** by looking at the residual plots of the fitted values against the predicted values.

From figure 13, we can clearly see that there is no pattern or trend with the points on the plots. This suggests that the linearity assumption for the link function holds.

From looking at these assumptions for Beta regression, we are able to say that some assumptions are met and some are not. For example, not all of the regions assume that the response variable is Beta distributed. This was evident through the histograms and Beta distribution curves for each individual region not following a similar shape. However, the response variable is assumed to be in the interval (0,1) due to the min-max scaling method that was used. Regarding the assumption for no multicollinearity, there was multicollinearity involved. However, when we removed predictors that were highly correlated with others, it led to the VIF values becoming worse. Lastly, the assumption of linearity for the link function was met, which was identified



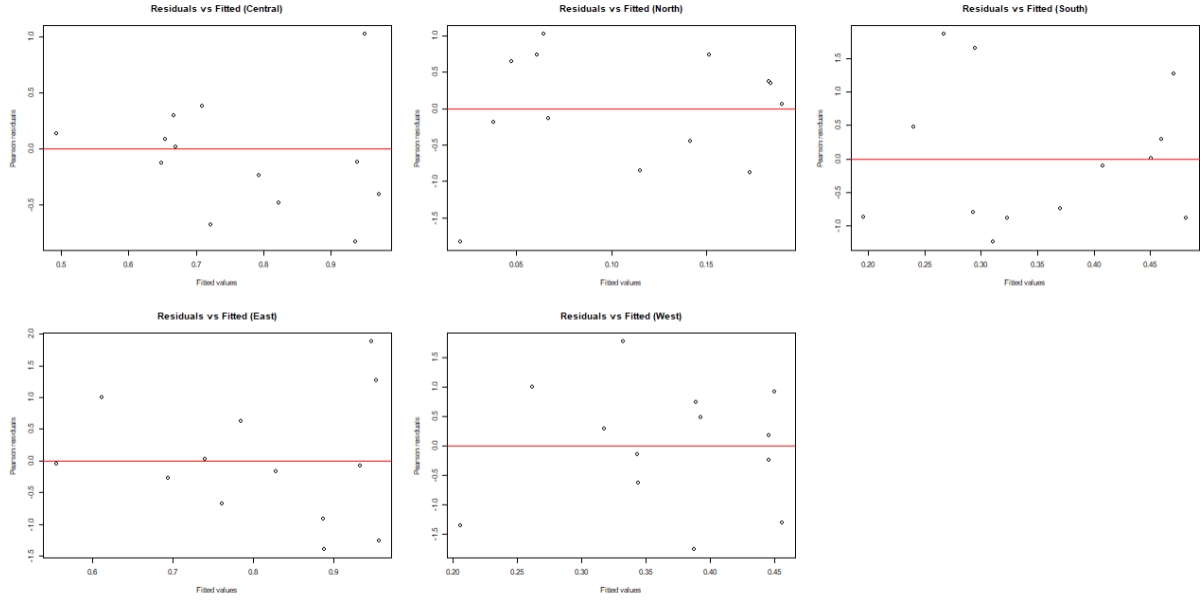


Figure 13: Residual plot for scaled Crime (By region)

through the residual plots showing no specific patterns.

## 5 Analysis of results and Discussion

This part of the research will delve into the results that we obtained through computing the different regression methods. It goes deeper into the different measures that we can look at to identify which models are seen to be more effective. We will look into some of the measures and how they are important to our objectives and analysis.

### Beta coefficients

Beta coefficients, that were provided through the different regression methods, identify which predictors were particularly significant for which regions. The interpretation of the beta coefficients varies based on which regression method was used. In general, for all methods, the beta coefficients show whether that specific predictor leads to an increase or decrease in the response variable. So, positive beta coefficients mean that the predictor causes crime to rise and likewise negative coefficients cause crime to fall. However, how can we tell by how much the crime response variable increases or decreases?

For Poisson and Negative Binomial regression, [Choueiry, 2021] interprets the beta coefficient as the expected change on the log scale in the response  $Y$  (due to the link function) by a unit change in  $X$ . Therefore,  $X$  increasing by 1 unit multiplies the rate of  $Y$  by  $e^\beta$  (known as the rate

ratio for counts). However, as Beta regression uses the logit function, the interpretations are different. The beta coefficients here represent the change within the log-odds of the response given a 1 unit increase in the predictor (known as odds ratio for proportions).

### Akaike Information Criterion (AIC)

The Akaike Information Criterion is a method used to evaluate which model fits best for the data that was used to generate the model. The best fit model is the one that uses the least amount of predictor variables to explain the greatest amount of variation.

### Statistical significance of the variables

Statistical significance is defined as when something has not happened due to chance or randomness. It helps quantify whether a result is likely due to chance or to some factor of interest [Gallo, 2016]. In our case, we use it to identify whether each predictor variable used in the model has a meaningful effect on the response variable. The measure we use to identify whether a predictor is significant is using the 'p-value'. The p-value shows the probability that the observed effect could've happened by random luck, assuming the null hypothesis to be true. If the p-value is less than the chosen alpha value, then there is sufficient evidence against the null hypothesis, which suggests that the variable is considered to be significant and has an effect. Whereas, a p-value greater than the chosen alpha value means that there is weak evidence against the null hypothesis, suggesting that the variable is insignificant and has no effect. In R, the different numbers of stars next to the p-value represent the level of significance for the 3 commonly used levels. Having more stars signifies having a smaller p-value. The standard alpha value generally used is  $\alpha = 0.05$  and that is what we will be using in the Poisson and Negative Binomial regression models.

To test whether the specific predictor is significant to the model, we test:

$H_0$  : Predictor  $x_i$  has **no relationship** with the number of crimes.

$H_1$  : Predictor  $x_i$  **does have relationship** with the number of crimes.

Through figure 14 obtained in [Hessing, 2020], we are able to identify which hypothesis needs to be rejected depending on the value of the p-value. As we are using 5% as the significance level, our  $\alpha = 0.05$ . Now, if the p-value is less than 0.05, then we reject  $H_0$  and assume that the specified predictor has a relationship with the crime number. Nevertheless, if the p-value is greater than 0.05, then we reject the alternative hypothesis ( $H_1$ ) and assume that the predictor has no relationship with the crime number.

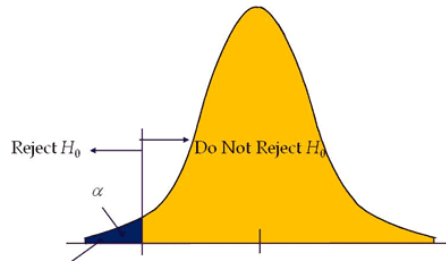


Figure 14: Left-tailed hypothesis test graph

## 5.1 Results from undertaking Poisson and Negative Binomial Regression

Table 6 shows the beta coefficients generated in R by using Poisson regression:

| Area    | Intercept  | Population<br>(In Mil-<br>lions) | Jobseeker's<br>Claimants<br>(%) | Median<br>House Price<br>(£) | Average<br>Income (£) | Job Den-<br>sity |
|---------|------------|----------------------------------|---------------------------------|------------------------------|-----------------------|------------------|
| Central | 13.3527631 | -0.6710625                       | 0.0368980                       | 0.0004540                    | -0.0301854            | 0.2058886        |
| North   | 13.18      | -1.641                           | 0.05061                         | 0.0009708                    | -0.02593              | 0.7223           |
| South   | 13.91      | -1.012                           | 0.05102                         | -0.0000496                   | -0.02412              | 0.194            |
| East    | 13.86      | -0.07314                         | -0.02064                        | -0.00008097                  | -0.03902              | -0.7811          |
| West    | 13.16      | 1.093                            | 0.02768                         | -0.00004308                  | -0.09248              | -1.310           |

Table 6:  $\beta$  values for predictor variables of each area using Poisson Regression

Table 7 shows the beta coefficients generated in R by using Negative Binomial regression.

### Interpreting results from Poisson and Negative Binomial regression

Question 2 being, 'Which socio-economic variables affect the number of crimes?' will be answered in this section of the study. Question 2a, 'What is the relationship between the socio-economic variables and the crime rate?' will be answered when the beta coefficients are interpreted and following on from that, question 2b, 'Which socio-economic variables are statistically significant predictors of crime?' will be also answered to identify which predictors influence crime in the different regions.

From table 6 and table 7, we can identify the general relationship between the predictors and the response for the two different count regression methods. One key thing to note from both tables

| Area    | Intercept  | Population<br>(In Mil-<br>lions) | Jobseeker's<br>Claimants<br>(%) | Median<br>House Price<br>(£) | Average<br>Income (£) | Job Den-<br>sity |
|---------|------------|----------------------------------|---------------------------------|------------------------------|-----------------------|------------------|
| Central | 13.3305846 | -0.6681343                       | 0.0354649                       | 0.0004483                    | -0.0302727            | 0.2221771        |
| North   | 13.1586667 | -1.5988773                       | 0.0489556                       | 0.0009177                    | -0.0258319            | 0.7015562        |
| South   | 13.8811650 | -1.0229303                       | 0.0515206                       | -0.0000875                   | -0.0231405            | 0.2385189        |
| East    | 13.87      | -0.07239                         | -0.02103                        | -0.00008997                  | -0.03899              | -0.7959          |
| West    | 13.09      | 1.207                            | 0.02568                         | -0.00002252                  | -0.09493              | -1.345           |

Table 7:  $\beta$  values for predictor variables of each area using Negative Binomial Regression

is that the values are very similar from both methods, and this will be visible when we look at the plots of the fitted models against the actual data points.

In the Central region of London, the Negative Binomial model performed better than the Poisson model, evident through the lower AIC value (263.03 against 982.7). This suggests overdispersion within the data for this region, and the Negative Binomial model handled it more effectively. From looking at the beta coefficients of the predictors, they suggest that population and income have a negative relationship with crime, suggesting that as population and income increase, the expected crime count falls. An increase in population means that there are more people out and about, especially in Central London due to the tourist attractions and west end, so these areas have higher public presence and increased surveillance, making it harder for individuals to commit crime. As the income of individuals increases, they are more likely to have economic stability, reducing the incentive to commit crime. However, when looking at jobseeker's claimants, house price and Job density, as they increase, they also lead to the expected crime count to increase, with house price having a weaker effect. Being unemployed and having to claim a benefit for it may lead to financial instability and may motivate individuals to engage in criminal activities. Expensive areas would attract property crimes such as burglaries and theft of expensive assets like cars. An increase in job density means more jobs available making areas busy because of workers commuting, which make crimes such as theft and pickpocketing easier. One key thing to note is that population and job density represent different dynamics. An increase in population represents stable residential presence but increase in Job density represents more workers commuting, leading to less monitored environments.

When looking at the statistical significance of the variables at the 5% level, the Poisson model

identifies all five predictors to significant with small p-values. However, the Negative Binomial model identifies all predictors except the house price predictor to be significant. Due to the overdispersion within the data, the Negative Binomial p-values are more reliable than those from the Poisson. Therefore, this suggests that for this model, house price predictor is insignificant (with a p-value of 0.14) meaning that there is 14% probability that the observed effect could have happened due to random chance.

In the North region of London, the Negative Binomial model again outperformed the Poisson model given the lower AIC (245.9 against 603.3). Looking at the beta coefficients for this region, all the predictors follow the same relationship as how they did for the Central region. However, the strength of the relationship differs. As stated before, these two methods interpret the beta coefficient as the expected change on the log scale in the response Y by a unit change in X. For example, looking into more detail for the jobseeker's claimants predictor from the Negative Binomial model, we look at the odds ratio ( $e^\beta = e^{0.049} = 1.0502$ ) that associates jobseeker's claimants to crime, which was 1.0502. This suggests that for each 1% increase in jobseeker's claimants, the expected crime rate increases by 5.02%.

Looking at the significance of the predictors for the North region, the Poisson model shows all five predictors to be significant. On the other hand, the Negative Binomial model identifies only the jobseeker's claimants and Job density predictors to be significant. Comparing to the central region, we see that jobseeker's claimants and job density are significant in both areas. However, the significance of predictors like income differ both regions. Income being significant in Central may suggest that uneven income distribution between wealthy and poor areas may play a key part. But, the North may have more even income distribution leading to the predictor to be insignificant.

For the South region of London, the Negative Binomial model was better compared to the Poisson model evident through the AIC values (250.01 against 577.22). When looking at the beta coefficients, all predictors followed the same relationship as how they did for the previous two regions, except for the house price now having a negative relationship with crime. This suggests that as the house prices increase, the crime rate falls. A possible reason for this could be that there is more security and surveillance placed within the region and also burglary rates are lower in the boroughs in this region compared to other regions. Looking at the change in log odds for income from the Negative Binomial model, the odds ratio ( $e^\beta = e^{-0.023} = 0.977$ ) suggests that for each unit increase in average income, the expected crime rate falls by 2.3%, with all other

predictors held constant. As incomes rise, the incentive to engage in crime falls due to people having stable employment and income.

When we look at the significance of the predictors for the South region, the Poisson model showed all predictors except house price to be significant. But, for the Negative Binomial model, the Population, jobseeker's claimants and income were significant. This suggests that for this region economic hardship and demographic pressure influence crime. However, unlike in the North, income plays a role in the South, which suggests a link between income inequality and crime. Comparing to the Central region, where all predictors except house prices are significant, it appears to be that the South region is influenced by specific socio-economic factors.

For the East region, the AIC values (240.09 against 317.05) suggest that the Negative Binomial model performed better than the Poisson model but not by a large amount like the other regions. When looking at the beta coefficients, the predictors follow a completely different relationship compared to the other regions discussed. All five predictors are seen to have a negative relationship with crime. So, as all the predictors increase, crime falls. This is quite a surprising result, but the significance of these predictors may provide more insight.

Looking at the significant predictors for the East region, the Poisson model identified all five of the predictors to be significant. However, when we look at the Negative Binomial model, all predictors were significant except the house price. Therefore, from the results, we identify that as the values of the predictors rise, the crime rate falls, which may reflect the initiatives taken within the community to keep the public safe or financial stability even though income levels are low for individuals. However, to explore the reasons behind the results of the beta coefficients, further research may need to be done to identify whether it is due to policy control, data artifacts (errors during collection) or different reasons.

Now looking at the last region, West of London, the Negative Binomial model once again performed better than the Poisson model, given the AIC values (254.55 against 733.40). The beta coefficients show that the population and jobseeker's claimants have a positive relationship with crime and the other three have a negative relationship. These relationships are similar to the East region, but now population and jobseeker's claimants have a positive relationship. As more individuals claim the unemployment benefit, the crime count increases. For all regions, an increase in income leads to fall in crime but the effect is strongest in the West region. This could possibly be due to greater income inequality, which makes the difference more impactful.

The statistical significance of the predictors from the Poisson model suggests that all predictors are significant except the house price (similar to the interpretation of the South model). However, the Negative Binomial model identifies the income and job density to be significant. The p-values for these predictors being extremely small suggest that these observed effects are most likely to not have happened due to random chance. Also, only these two predictors being significant means that the East region is influenced particularly by these two predictors.

We can now go into more detail regarding the significant predictors from the Negative Binomial models by looking at the interpretations of the p-values. We will be looking at the 5% significance level. For the **Central part of London** model, all predictor variables were significant except from the house price. The insignificant house price predictor has the largest p-value (0.14) out of all five p-values meaning that there is 14% probability that the observed effect could have happened due to random chance. Looking at the model for **North of London**, the results show that the only significant predictors are jobseeker's claimants and Job density. Having p-values of 0.006 and 0.010 respectively suggests that there is a 0.6% probability that the observed effect could have happened due to chance for jobseeker's claimants. There is a 1% probability for the job density predictors effect to have happened due to chance. These probabilities are quite small, which is evidence for both predictors to be significant in this model. On the other hand, this suggests that the rest of the predictors are not significant to this particular model due to the insufficient evidence against the null hypothesis. In the model for **South of London**, the house price and job density predictors are insignificant. They have p-values of 0.817 and 0.581 respectively, showing that the observed effect is likely to happen due to chance (81.7% and 58.1% probability of effect being due to chance). Therefore, the other three predictors are significant within this model. For the **East of London** model, all of the predictors are significant except from the house price. The p-value for the house price predictor is 0.447. So, there is 44.7% probability that the observed effect is due to chance. The p-values for the other four predictors are very small showing that they are significant. Lastly, we look at the **West of London** model and see that the only two significant predictors are average income and job density with extremely small p-values. Hence, there is insufficient evidence to say that the remaining four predictors have a significant effect on the crime response variable for this particular model.

After all these assumptions of identifying which variables were significant for which models, one key point to notice is that for the Negative Binomial regression models, there were fewer significant predictors compared to what was obtained from the corresponding Poisson regression

models. A possible reason for this is that one of the assumptions for Poisson regression is for the mean and variance to be equal. However, if there is overdispersion, then the Poisson model will underestimate the variability within the data. The standard errors measure how well the model fits the data by obtaining the difference in the observed and predicted values and identify how accurately the model estimates its coefficients. Due to the overdispersion and underestimating, the model is more confident about the beta coefficients than it should be, resulting in smaller standard errors. This will then lead to inflated test statistics and artificially small p-values. The 'inflated test statistics' occur because the model is underestimating the standard errors, as it assumes that variability will be low. This results in test statistics being larger than what they actually should be and gives the false impression that the predictors have strong effects. Due to this, the test statistics are large and the p-values become extremely small. These small p-values make it look like the predictors are significant when they may not be and could lead to incorrect conclusions. Therefore, the assumption of mean and variance being equal is important, and if there is overdispersion and is not accounted for, then it may lead to underestimating significant variables and will lead to misleading results.

However, having a less number of significant predictors being obtained from Negative Binomial regression gives a useful insight into the model choice we made. We recall from earlier that the Negative Binomial regression has a dispersion parameter ( $\theta$ ) that represents the overdispersion. This leads to large standard errors as the Negative Binomial regression corrects the underestimation from the Poisson regression. The large standard errors mean that the model is less confident about the beta coefficients. As a result of this, the test statistic is smaller, but the p-value is larger. However, these p-values are now more realistic because they represent the actual variability within the data and overcome the issue of the overdispersion within the Poisson model.

## 5.2 Results from undertaking Beta Regression

Table 8 shows the beta coefficients generated in R by using beta regression:

### Interpreting results from Beta regression

From table 8, we obtain the beta coefficients of the predictors from using Beta regression. When we look at the beta coefficients at first glance, the relationship looks similar to those coefficients obtained in the Poisson and Negative Binomial models. However, there are some regions with predictors that have opposite relationships compared to those obtained by the count models.



| Area    | Intercept | Population<br>(In Mil-<br>lions) | Jobseeker's<br>Claimants<br>(%) | Median<br>House<br>Price (£) | Average<br>Income (£) | Job Den-<br>sity |
|---------|-----------|----------------------------------|---------------------------------|------------------------------|-----------------------|------------------|
| Central | 4.400216  | -2.612073                        | 0.415730                        | 0.004510                     | -0.312174             | 3.210080         |
| North   | 11.997479 | -14.856392                       | 0.336182                        | 0.009421                     | -0.131507             | 3.594092         |
| South   | 7.8035253 | -4.3135438                       | 0.1982960                       | -0.0001099                   | -0.0884699            | 0.4899922        |
| East    | 13.210192 | 0.730105                         | -0.064079                       | -0.007801                    | -0.401113             | -5.096079        |
| West    | 4.3180279 | 4.3410416                        | 0.1011827                       | 0.0001701                    | -0.3625772            | -5.2811053       |

Table 8:  $\beta$  values for predictor variables of each area using beta Regression

Looking at the beta coefficients for the Central model, the relationships are the same as what was obtained by the count regression models. However, the strength of the effects differ, particularly for the population and job density predictors. Looking at the odds ratio for the population and job density predictor, we get 0.073 ( $e^\beta = e^{-2.61} = 0.073$ ) and 24.8 ( $e^\beta = e^{3.21} = 24.8$ ) respectively. We can interpret this values separately for each predictor. A one unit increase in population reduces the odds of a higher crime rate by roughly 92.7%, holding all other predictors constant. However, a one unit increase in job density increases the odds of a higher crime rate by 24.8%. When we look at the significance of these predictors, we see that none of the predictors are significant at the 5% level. This suggests that for the Central region, there are no specific factors that influence crime.

For the North model, the beta coefficients once again show a similar relationship but the strength of the effects differ by quite a large amount here, particularly for the population predictor. The beta coefficient for the population predictor is -14.856 and when we compute the odds ratio, we obtain a very tiny value of 0.000000353. From this, we can conclude that a one unit increase in the population reduces the odds of a higher crime rate by around 99.9%. This tells us that increasing population in this area leads to extremely less likeliness of having high crime rates, holding all other variables constant. When looking at significance for this North model, we see that only the population and jobseeker's claimants predictors are significant here. This suggests to us that the crime in the North region is influenced by these specific socio-economic factors.

For the South model, the beta coefficients represent similar relationships. For example, the jobseeker's claimants and job density predictors have a positive relationship meaning that as they

increase, crime also rises. On the other hand, population, house price and income have a negative relationship leading us to assume that as these predictors increase, the crime rate falls. Once again, the population predictor has a stronger effect within this model suggesting that it plays a big part in leading to the crime rate to fall. The significant predictors within this South model are population, jobseeker's claimants and job density. These factors suggest that for the South region demographic size, distribution of employment and unemployment pressure are factors that have a key influence on crime trends in this area.

Looking at the East region model, the beta coefficients have an identical relationship to the count model beta coefficients, but here the population coefficient now has a negative relationship with crime. This suggests that as the population rises in this region, the crime rate falls. When looking at the significance of the predictors, the only predictor seen to be significant is the income. This suggests that the crime patterns in this area are strongly influenced solely by this predictor.

Lastly, looking at the beta coefficients for the West region model, we see that the relationships of the predictors and response are similar to those obtained from the previous two methods, but the house price predictor has a positive relationship. However, as the coefficient of the house price predictor is extremely small, it suggests that it does not have much of any impact within the model. The significant predictors for the West model are income and job density. Income, being significant in this model, suggests that uneven distribution of income between the poor and rich areas is an important factor within this area.

Therefore, by looking at the beta coefficients obtained through beta regression, we identify that most predictors have a similar relationship to crime, as those obtained from the count models. One key difference between the count regression methods and the Beta regression method is that the link functions. So, in Beta regression, the coefficients represent the change in the log-odds of the mean crime proportion, given a one unit increase in the predictor. There are similar relationships to the count models, which suggest that those specific predictors are constantly influencing crime regardless of whether we are modelling count or proportion. For example, the Income predictor has negative beta coefficients for all regions and all methods, which strengthens the argument of an increase in income being associated with lower crime.

To conclude this section, the comparative analysis of the beta coefficients obtained through the Poisson, Negative Binomial and Beta regression models have provided a key insight into the significant socio-economic factors that play a part within the trends of crime in the five differ-

ent regions. All three models identified certain predictors to be significant across most regions such as Population, Job density, and jobseeker's claimants.

When we looked at the beta coefficients and significance of the predictors obtained through Poisson regression, we identified that the method reported more significant predictors due to overdispersion. This was better accounted for by the Negative Binomial model, which provided more accurate p-values, due to the model having a specific parameter for overdispersion. The Beta regression modelled crime as a proportion. The results obtained were very similar to those obtained through using the count-based models but highlighted some different relationships for certain predictors.

These differences within the results emphasise the importance of choosing models based on the data within regression analysis. Each of the models that were used have particular individual strengths and we were able to identify the strength of the Negative Binomial model when the results were shown to be more reliable, due to the overdispersion being considered.

### 5.3 Model equations of the fitted regression models

This section of study will look into the model equation of the different regression methods used. The predictor variables for all of the equations are denoted by the following:

$x_1$  - Population (in millions),  $x_2$  - jobseeker's Claimants (%),  $x_3$  - Median house price (in thousands),  $x_4$  - Average income (in thousands),  $x_5$  - Job density.

#### Model equations from Poisson regression

Firstly, we look at the model equations obtained through undertaking Poisson regression. *Note that  $Y_i \sim \text{Poisson}(\lambda_i)$ .*

$$\text{Central: } \log(\lambda) = 13.35 - 0.67x_1 + 0.04x_2 + 0.0005x_3 - 0.03x_4 + 0.21x_5$$

$$\text{North: } \log(\lambda) = 13.18 - 1.64x_1 + 0.05x_2 + 0.001x_3 - 0.03x_4 + 0.72x_5$$

$$\text{South: } \log(\lambda) = 13.91 - 1.01x_1 + 0.05x_2 - 0.00005x_3 - 0.02x_4 + 0.19x_5$$

$$\text{East: } \log(\lambda) = 13.86 - 0.07x_1 - 0.02x_2 - 0.00008x_3 - 0.04x_4 - 0.78x_5$$

$$\text{West: } \log(\lambda) = 13.16 + 1.09x_1 + 0.03x_2 - 0.00004x_3 - 0.09x_4 - 1.31x_5$$

From looking at the model equations obtained through Poisson regression, we are able to identify which predictors have a stronger or weaker effect by comparing the five regions. When looking at the population predictor, we see that all areas except from the West have a negative

relationship with crime. The strength is the strongest in the North, suggesting that this predictor is more impactful in this certain region. For the jobseeker's predictor, the relationships vary depending on the regions. However, the effects are similar for all regions implying that this predictor had an equal influence on crime for all regions. For the house price predictor, we see that the effect is weaker in most regions except from the North. From this, we can assume that this specific predictor has a key influence mainly in the North. For the income predictor, there is a negative relationship with crime for all regions but the strength is slightly stronger in the West. This may suggest larger income disparities within the West. Lastly, the job density predictor has strong effects in the North and East. As the job density rises in the East, crime falls.

### Model equations from Negative Binomial regression

Next, we look at the equations derived through Negative Binomial regression. *Note here that  $Y_i \sim NB(\mu_i, \theta)$ .*

$$\text{Central: } \log(\lambda) = 13.33 - 0.67x_1 + 0.04x_2 + 0.0004x_3 - 0.03x_4 + 0.22x_5$$

$$\text{North: } \log(\lambda) = 13.16 - 1.60x_1 + 0.05x_2 + 0.0009x_3 - 0.03x_4 + 0.70x_5$$

$$\text{South: } \log(\lambda) = 13.88 - 1.02x_1 + 0.05x_2 - 0.00009x_3 - 0.02x_4 + 0.24x_5$$

$$\text{East: } \log(\lambda) = 13.87 - 0.07x_1 - 0.02x_2 - 0.00009x_3 - 0.04x_4 - 0.80x_5$$

$$\text{West: } \log(\lambda) = 13.09 + 1.21x_1 + 0.03x_2 - 0.00002x_3 - 0.09x_4 - 1.35x_5$$

From looking at the model equations that were obtained through computing Negative Binomial regression, we see that the relationships between the predictor variables and crime are the same as those calculated from the Poisson models. Here, there are small changes within the effects of the predictors, which suggests no major differences from the Poisson models. The consistency in the coefficients from both models increases our confidence in these findings.

### Model equations from Beta regression

Lastly, we look at the equations derived through Beta regression. *Note here that  $Y_i \sim \text{Beta}(\mu_i, \phi)$ .*

$$\text{Central: } \log(\lambda) = 4.40 - 2.61x_1 + 0.42x_2 + 0.005x_3 - 0.31x_4 + 3.21x_5$$

$$\text{North: } \log(\lambda) = 12.00 - 14.85x_1 + 0.34x_2 + 0.009x_3 - 0.13x_4 + 3.59x_5$$

$$\text{South: } \log(\lambda) = 7.80 - 4.31x_1 + 0.20x_2 - 0.0001x_3 - 0.09x_4 + 0.49x_5$$

$$\text{East: } \log(\lambda) = 13.21 + 0.73x_1 - 0.06x_2 - 0.008x_3 - 0.40x_4 - 5.10x_5$$

$$\text{West: } \log(\lambda) = 4.32 + 4.34x_1 + 0.10x_2 + 0.0002x_3 - 0.36x_4 - 5.28x_5$$

Through looking at the model equations derived by Beta regression, we see that the relationship changes for some variables and the strength of the effects also vary. When looking at the

population predictor, we see that the strength of the effects are quite stronger compared to those obtained from the count regression models. There is a stronger effect of negative relationship with the population predictor in the North, suggesting that this predictor is particularly important within this region by a significant amount. For the jobseeker's claimants, house price and income, the effect of the predictors on the crime are quite similar for all regions. However, when we look at the effect of the job density, we see that the effects are stronger than those obtained from the count regression models. In particular, this predictor has a strong influence on crime in all regions except from the South. One of the main reasons for the differences in the results could be due to this regression method modelling proportion-based crime data and uses the logit link function instead of the log link function for the response variable.

### **Region Interpretation using Negative Binomial model**

In order to answer question 3, 'Are there significant differences in the crime rate between the various regions after considering the socio-economic variables?', Negative Binomial models was used to control the overdispersion. To identify whether there were variations in crime rates across regions, the two following models were compared using the likelihood ratio test (with the Central region as the 'reference' category):

$$\begin{aligned} \log(\lambda) &= \text{Region} + \text{Population}x_1 + \text{jobseeker's}x_2 + \text{Houseprice}x_3 + \text{Income}x_4 + \text{Jobdensity}x_5 \\ \log(\lambda) &= \text{Population}x_1 + \text{jobseeker's}x_2 + \text{Houseprice}x_3 + \text{Income}x_4 + \text{Jobdensity}x_5 \end{aligned}$$

From looking at the results, the North, South and West have a negative beta coefficient suggesting that these three regions have a lower crime rate compared to the Central region. However, the East region has a higher crime rate. When looking at the significance, the North region had a significantly lower crime rate compared to the Central region, due to the extremely small p-value (0.00000118). The East and West regions also had a significantly lower crime rate compared to the Central region. However, the South showed no significant difference.

The likelihood ratio test comparing the models with and without the 'Region' variable showed a statistically significant improvement in model fit when region was included ( $p < 0.05$ ). This answers question 3 by stating that even after considering the the various socio-economic indicators, the differences in regions play a vital role in explaining the variation of crime trends across London.

### **Assessing the impact of the 2008 financial crisis**

The financial crisis in 2008 was a major event which led to a global economic recession due to

the collapse of the US housing market and the failure of financial institutions. This crisis had an impact on the UK by causing a deep recession, which led to increased unemployment and many businesses failing, leading to overall confidence and uncertainty within the UK economy. The sharp increase in unemployment left people in a difficult situation, especially income instability as there were not many jobs available. This phase may have incentivised individuals to commit criminal activities to gain wealth and assets for themselves by taking them away from others.

To answer question 4, 'Did the 2008 financial crisis lead to vast changes in crime rates across London?', a dummy variable was created for the year predictor to identify the impacts after the financial crisis. The model equation using Negative Binomial regression and including the crisis variable provided the following insights. When looking at the impact of this event across the whole of London, the negative beta coefficient ( $-0.05$ ) suggested there was a slight fall in crime during and after the crisis. However, when looking at the p-value ( $0.169$ ), this is not statistically significant and therefore leads us to assume that we cannot for sure say that the crisis had a strong impact on crime across London.

However, when diving deeper into the regional impacts of the financial crisis, we can answer question 4a, 'Were the impacts greater in some regions compared to others?' by looking at the individual Negative Binomial models with the crisis variable. The Central and West region saw a small drop in the crime rate. Central had a beta coefficient of  $-0.02$  but was not significant due to the p-value of  $0.22$ . West had an even smaller coefficient of  $-0.00074$ , but was not significant due to the p-value of  $0.79$ . On the other hand, when we look at the other three regions, we see that the beta coefficients are positive but are only significant for the North and East region. From these results, we can assume that the main regions that the crisis had an significant impact on were the North and East regions. These regions experienced an increase in crime, possibly due to the increasing unemployment or financial instability of individuals.

To answer question 4b, 'Are higher levels of jobseeker's claimants associated with higher crime rates during and after the financial crisis?', an interaction term between jobseeker's claimants and crisis indicator was used. This interaction term represents how the effect of jobseeker's claimants on crime changed during/after the 2008 financial crisis compared to before. This interaction term was included in each regional model. The results gave us insight into the crime variations between regions as a result of the interaction term. In the Central and West region, the interaction term was negative and statistically significant (Central:  $\beta = -0.10$ , p-value =  $0.01$ ; West:  $\beta = -0.099$ , p-value =  $0.0097$ ). From this we can identify that the positive re-

lationship between crime and jobseeker's claimants slowly weakened after the financial crisis. For the South region, the beta coefficient was negative but insignificant. For the North and East region, the interaction terms were also insignificant (North:  $\beta = -0.04$ , p-value = 0.38; East:  $\beta = 0.017$ , p-value = 0.32). This suggests that the crisis did not affect the relationship between jobseeker's claimants and crime for these regions.

### **Fitting models using interaction terms**

In order to answer question 5, 'Do interactions between socio-economic variables (e.g. Job Density and Average Income) give a better insight into crime patterns compared to individual factors being analysed alone?', Negative Binomial models were fit for each region, which included an interaction between the job density and average income predictors. In many areas, the combination of job density (which is indicator for economic activity) and income (indicator of economic status) could influence crime differently. In wealthier areas, greater availability of jobs may reduce crime due to individuals having higher confidence in being employed. But in poorer areas, job density could lead to an increase in crime because of higher competition for the limited amount of available jobs. This may lead to disputes between people if they are unsuccessful, which may then further escalate. Also, if individuals see that they will never be able to get hired for these jobs, they may give up and choose wrong paths of earning money easily such as robbery, drugs and various other harmful methods.

When looking at the results of the models that included the interaction term, the findings showed that none of the regions showed the interaction term to be significant (p-value > 0.05). When the original Negative Binomial model was compared to the model with the interaction term using the AIC values, there was no meaningful improvement in including the interaction term. These results suggests that for this dataset, individual factors like income and job density act independently in their relationship to crime, rather than interactively.

### **Fitting the regression models to the original data**

The last thing that was done to identify which model of the three was the best fit to the data was looking at the plot of the original data points and fitting the different models to these data points. Figure 15 and 16 show the plots of the regression models being fitted to the original data points. Looking at figure 15 at first glance, we see that the Negative Binomial and Poisson regression models produced fitted curves, which are both closely overlapped when plotted against the actual data points. This shows that both models are capturing the same trend in the crime rates. This is mainly due to the fact that they both use the same link function and they share the same

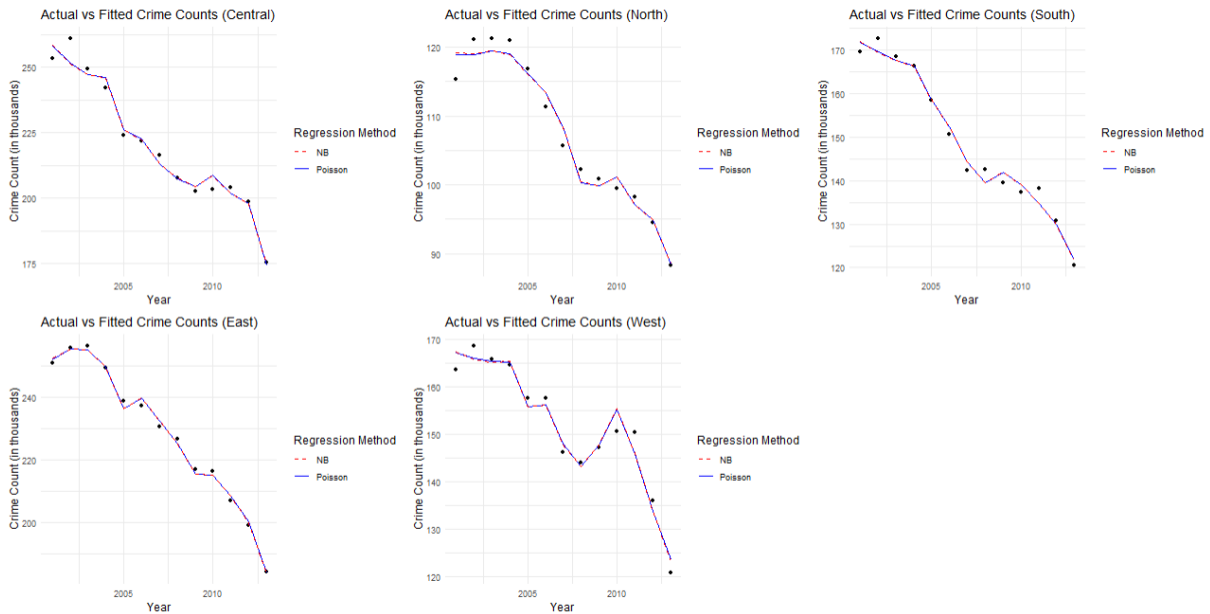


Figure 15: Fitting Poisson and Negative Binomial regression models to the data (By region)

mean structure. However, the overlap of both models does not mean that they are both suitable. We know that the dataset involves overdispersion and the Negative Binomial model accounts for this, but the Poisson model assumes equidispersion. Therefore, choosing the Negative Binomial model is a more effective choice, especially when looking at the significance of predictors. When looking at the plots, we see that the models fit the points quite nicely.

However, when we look at figure 16, this plot represents the beta models being fitted to the data for the different regions. We see that the model looks like it fits the data points, but we see that the Negative Binomial models fit the data points better as they go through majority of the points. But, when we look at the Beta model being fit, we see that it misses quite a few points even though it follows the same shape as the lines in figure 15.

## 6 Conclusion and Future Work

### 6.1 Conclusion

This research paper explored into the crime patterns in London between 2001 and 2013, with a key focus on the socio-economic factors that influence crime and the variations in crime trends between the five different regions. Through the use of multiple regression methods (Poisson, Negative Binomial, and Beta regression) and various descriptive analysis methods, the analysis conducted within this paper addressed a combination of research questions based on crime trends over time, significance of predictors, and impact of external events (2008 financial crisis).



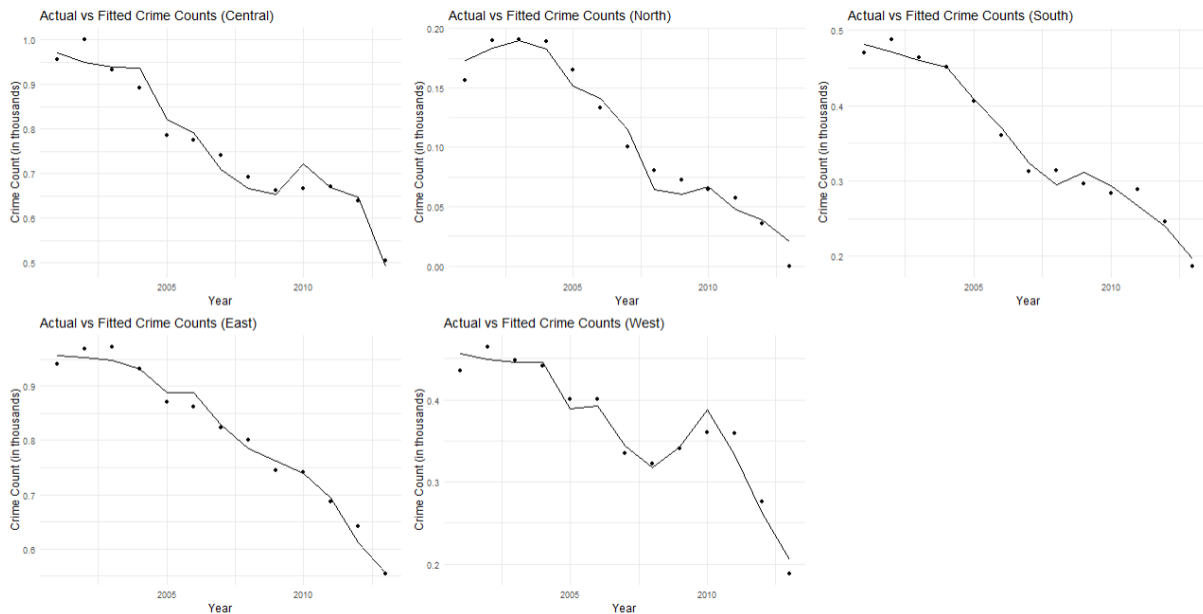


Figure 16: Fitting Beta regression model to the data (By region)

To answer question 1 regarding the general trend of the crime, the plot of the crime over the years from section 4.1 showed the general trends and fluctuations for the various regions. It was obvious to see that the crime rates fell over time across all regions but there were certain periods where there were sharp rises and falls for certain regions. Possible reasons for this could be regional differences and region-based events, such as the 2005 attacks in the Central areas of London leading to lower crime rates during that period. Also, when looking at the West region, it could be assumed that the 2008 financial crisis had a greater influence within that region, possibly leading to mass amounts of unemployment. So, the potential key drivers for the fluctuations in the crime rate could be due to the temporal events or various other socio-economic factors, which differ according to region.

There were many socio-economic variables considered which potentially influenced crime. However, when regression analysis of the data was computed, the results showed particular socio-economic factors that directly influenced crime for that region. Some of these factors were seen to be significant to affecting the crime rate but some were identified to have no impact on the crime rate. When undertaking the regression methods, it was identified that the Negative Binomial regression models were a more accurate and reliable fit due to the overdispersion being controlled. Therefore, the Negative Binomial model provided more reliable p-values. Whereas, the Poisson models assumed equidispersion, which underestimated the variability within the data and provided inflated test statistic and smaller p-values. When looking at the significance of the predictors, different predictors were significant for different regions. For example, in the Central region, all predictors except from median house price were seen to be significant. This

suggests that within this Central region, those four specific predictors play a part in influencing the crime rate. However, if we look at the West region, the average income and Job density are significant, which suggests that there may be income inequality within this area and there could be a limited amount of job available for individuals meaning that they may turn to criminal methods. Thus, we see these two predictors to be significant in this West region and other regions have other specific predictors that affect the crime rate within that area. From this insight, we identify specific predictors to be significant to crime depending on the region we are analysing.

To identify whether crime rates varied from region to region, a model was assessed, which included the 'region' as a predictor. From the results, we saw that when we were comparing to the Central region, the East region had a higher crime rate but the other three regions did not. When testing the models using the likelihood ratio test, we saw that the model that included the region term was a better fit. This suggested that each individual region had different crime trends compared to other region. These crime trends were affected by the socio-economic factors for that region and these factors vary region by region. Some regions may see certain factors as a negative as it may influence the crime rate negatively by a significant amount. However, other regions may see that same factor as a positive and it may influence the rate in a positive way.

Looking at the impact of the 2008 financial crisis, this crisis led to a sharp increase in unemployment and a expected increase in jobseeker's claimants across all boroughs and regions. The crisis had affect the UK economy by leading it to an economic recession. However, some areas of London were hit more harder than others. For example, the North and the East regions suffered from the financial crisis more than the other three regions. North of London did not directly get impacted during the years of the financial crisis but as a result of the crisis, the region suffered from long-term effects. House prices continued to rise in the North but wages slowly began to stagnate, making the area less popular to live in. Similarly, for the East of London, many boroughs took a long time to recover from the impacts and the East boroughs began to grow and recover after long periods of time compared to the other part of London.

To identify which factors had a direct impact on the crime rate, many regression models were tested using individual terms. However, to identify whether interactions of factors had a greater significance, the interaction term between income and job density was tested. The results showed the interaction term to be insignificant for all regions. This suggested that individual factors are more likely to have a relationship and impact with the crime rate compared to interaction between different factors.

To conclude with, throughout the analysis and computations, it was quite evident that different regions had different predictors that were significant to the crime rate. When looking at the crime hotspots within London from section 4.1, it was clear that various boroughs had different crime rates and these crime rates changed over a period of time. However, some of the key influencers for the crime rate to change over the 13 year period were the five predictors that were used in this research. Through the regression analysis, we worked out that for some regions the predictors influenced the crime rate positively by decreasing it. However, for other regions, these same predictors influenced the crime rate in the opposite way by leading it to rise. As an example, average incomes were quite high in the Central part of London, which led to economic stability and less incentive to engage in crime. However, in the East part of London, average incomes were significantly lower, which may push individuals to engage in criminal methods to gain more wealth. These predictors vary region to region and are affected by certain living standard and lifestyles depending on the different regions.

## **6.2 Recommendations**

Given the results that were obtained in section 5.1, 'Results from undertaking Poisson and Negative Binomial regression' and section 5.2, 'Results from undertaking Beta Regression', I would have liked to consider a wider range of factors, such as literacy rate, ethnic diversity, poverty and mental health statistics. These additional factors could provide more insight into crime trends and would help offer more actionable insights for targeted interventions to help reduce crime across London.

Secondly, I believe expanding the geographical scope beyond London or even possibly beyond the UK may help identify whether the patterns observed in London are consistent nationwide or are restricted just to the London boroughs. Analysing crime trends and potential factors that influences crime for different countries, like UAE, may provide key insights on policies that may need to be implemented within London to help reduce the crime rate. Through analysing crime in different countries, we may be able to identify the key factors influencing crime and how the governing bodies tackled the issue of rising crime.

Lastly, I believe that more recent data (maybe post 2013) would provide more up-to-date patterns of crime. In recent times, there have been many social, economic and political developments and changes, which may have determined changes in crime patterns. Covid-19 and Brexit were two recent key events, which may have had an impact on the socio-economic conditions and

crime patterns across London and the UK.

Overall, improving on this study through more recent datasets, greater geographical scope, and more variety of variables may lead to more informed policy-making strategies and more crime prevention methods, which are specifically tailored to the needs of the different regions.

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